ELEC3705 Lecture Notes

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1 Why Quantum Mechanics?

1.1 Photoelectric Effect

The photoelectric effect allows us to extract electrons from a piece of metal using light/high-powered lasers.

1. Below a certain frequency (colour) of light, no electrons are extracted from the metal. This is completely independent of the light intensity.
2. Above a cutoff frequency ω_c, the electrons are extracted from the metal, even at very small light intensities.
3. The energy E of the extracted electrons is proportional to $\omega - \omega_c$.
4. The momentum $p = mv = \sqrt{2mE}$ of the extracted electrons is inversely proportional to the wavelength of light.

1.2 Planck-Einstein Relations

This effect can be explained is we assume that light in quantised in packets of energy, called photons, where

$$E = \hbar\omega = hv$$
$$\vec{p} = \hbar\vec{k} = h/\lambda,$$

where $h = 6.626 \times 10^{-34}$J s is the Planck constant, and $\hbar = h/2\pi = 1.054 \times 10^{-34}$J s/rad is the "reduced" Planck constant. These are the Planck Einstein relations. They describe light as composed of "particles" of energy, determined by the frequency.

1.3 Double-Slit Experiment

The double-slit experiment demonstrated that light is also an electromagnetic-wave, and can be understood by using the linearity of Maxwell's equations.

The light intensity is proportional to $|E|^2$, and if $\vec{E}(\vec{r}) = \vec{E}_1(\vec{r}) + \vec{E}_2(\vec{r})$ from the two sources, then

$$I \propto \left|\vec{E}_1(\vec{r}) + \vec{E}_2(\vec{r})\right|^2 \neq \left|\vec{E}_1(\vec{r})\right|^2 + \left|\vec{E}_2(\vec{r})\right|^2,$$

shows the wave nature of light.

This experiment still holds at extremely low intensities. It is still individual photons that show up on the screen, however when averaged over time it still results in the interference pattern.

1.4 Double-Slit with Particles

This experiment can be repeated using things that are normally thought of as particles, such as electron or neutrons. To find the required wavelength, we can use the de Broglie hypothesis

Definition 1.1. The de Broglie hypothesis states:
"to every particle is associated a wave, which obeys the same Planck-Einstein relations as applicable for photons."

From the relation above, we can derive

$$\boxed{\lambda = \frac{2\pi}{k} = \frac{h}{p}}$$

This is only significant for extremely small objects. The distance between intensity peaks is given by

$$\delta x = \lambda \frac{D}{d},$$

with $\lambda = h/p$, D is the distance between the slits and the screen, and d is the separation of the slits. This means that larger particles have smaller distances between intensity peaks, with all else being the same. To make a meaningful observation, we can

- Increase the wavelength by decreasing the momentum (smaller velocity for constant mass),
- Decrease the distance between the slits,
- Or increase the distance to the screen.

1.5 Wave Function

These two experiments force us to recognise the idea that particles are simultaneously particles and waves, and in fact that waves and particles are not different things. Instead, each wave is made of packets of energy, with is associated with wavelength $\lambda = h/p$ and frequency $v = E/h$.

Instead of trying to describe the trajectory of a particle with position $\vec{r}$ and momentum $\vec{p}$, we use a wave function $\psi(\vec{r}, t)$, which contains all the information we can gather on the particle. This wave function is a probability amplitude, such that

$$C \iiint_V |\psi(\vec{r}, t)|^2 d^3\vec{r} = P_V(t),$$

where $P_V(t)$ is the probability of finding the particle in the volume V at time t, and C is a normalisation constant to ensure the value is between 0 and 1, i.e.

$$C \iiint_{-\infty}^{\infty} |\psi(\vec{r}, t)|^2 d^3\vec{r} = 1,$$

i.e. the particle has to be somewhere. Now instead of calculating the trajectory of a particle, we instead calculate the probability of finding the particle in a certain region in space, and how that evolves over time.

We know that Maxwell's equations allow solutions in the form of propagating waves, e.g.

$$\vec{E}(\vec{r}, t) = Ae^{i(k\vec{r} - \omega t)}$$

is a plane wave that solve the wave equation of the form

$$\nabla^2 \vec{E} = \frac{1}{v^2} \frac{\partial^2 \vec{E}}{\partial t^2},$$

where $v = \omega/k$. This wave equation contains the partial second derivative of the electric field with respect to both time and space. Taking the second derivative of the plane wave results in the coefficients k^2 and ω^2 which give rise to the $1/v^2$ term in the equation. Note that v is constant.

Consider a free particle. The energy is purely kinetic, so

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m},$$

which we can impose Planck-Einstein to get

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}.$$

An important thing to note is that the energy depends quadratically on k, but linearly on ω. This means that an appropriate wave equation should contain a first derivative with respect to time, and the second derivative with respect to space. Here, v is no longer constant.

1.6 Schrödinger Equation

Schrödinger postulated an equation that describes the time evolution of the wavefunction of quantum object to be

$$\boxed{i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r},t) + V(\vec{r})\psi(\vec{r},t)}$$

$V(\vec{r})$ is a (position-dependent) potential energy. It is similar to a wave equation, containing the first derivative w.r.t. time, and the second w.r.t. space. The coefficients are chosen to impose the Planck-Einstein relations. Because of the first derivative w.r.t. time, the equation explicitly contains i.

Proof. Consider a free particle not subjected to any potentials or forces ($V(\vec{r}) = 0$):

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r},t).$$

This has a solution of the form

$$\psi(\vec{r},t) = Ae^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

with the condition (see Problem 1.4 for a proof of this condition)

$$\hbar\omega = \frac{\hbar^2k^2}{2m}.$$

Inserting the Planck-Einstein relations, the above conditions means

$$E = \frac{p^2}{2m} = \frac{1}{2}mv^2.$$

□

1.7 Wave Packets

These wavefunctions can be combined into a single wave packet by simply summing them together (and hence with 3 wave numbers k). The interference between the different wave components causes the wave components to no longer be spread out equally in space, however it still have recurrences.

These recurrences get more spread out the more wave components are added, up to a limit of infinitely many components, each with a different wave number k and with a coefficient $g(k)$. For simplicity, take the 1D case ($\vec{r} \equiv x$):

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g(k)e^{i(kx-\omega t)}dk.$$

We can find $g(k)$ by setting $t = 0$ (i.e. initial conditions)

$$\psi(x,0) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g(k)e^{ikx}dk,$$

therefore

$$g(k) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \psi(x,0)e^{-ikx}dx.$$

This means that $g(k)$ is the Fourier transform (in space) of the shape of the wave packet at $t = 0$, i.e. of the probability amplitude of finding the object at point x at $t = 0$.

1.8 Heisenberg Uncertainty Principle

Consider a Gaussian wave packet

$$\psi(x,0) = \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{x^2}{2\sigma_x^2}\right).$$

It's Fourier transform is also a Gaussian

$$\begin{aligned} g(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x,0) e^{-ikx} dx \\ &= \frac{1}{\sqrt{2\pi}\sigma_x} \exp\left(-\frac{k^2}{2\omega_k^2}\right). \end{aligned}$$

From the properties of Fourier transforms, we know that

$$\sigma_k = 1/\sigma_x \quad \Rightarrow \quad \sigma_k \sigma_x = 1.$$

Imposing the Planck-Einstein relation $p = \hbar k \Rightarrow \sigma_p = \hbar \sigma_k$, we get

$$\boxed{\sigma_p \sigma_x = \hbar}$$

This relation is a simple mathematical property of Fourier transform. This appears as soon as you assume the Planck-Einstein relations, i.e. wave-particle duality. This also holds if you assume the uncertainty principle as a postulate, then wave-particle duality is a consequence of the mathematical properties of Fourier transforms.

1.9 Problems

Problem 1.1. A particle of mass $1ng$ moves at $1mm/s$. Estimate its de Broglie wavelength and compare it to atomic dimensions

Answer:

The de Broglie hypothesis states that the wavelength of an object is given by

$$\lambda = \frac{2\pi}{k} = \frac{h}{p},$$

where λ is the wavelength, k is the wave vector, h is the Planck constant, and p is the classical momentum. Plugging in our values we get

$$\begin{aligned} p &= 1 \times 10^{-9} \times 1 \times 10^{-3} \\ &= 1 \times 10^{-12} \\ \Rightarrow \lambda &= \frac{h}{p} \\ &= \frac{6.626 \times 10^{-34}}{1 \times 10^{-12}} \\ &= 6.626 \times 10^{-22} m. \end{aligned}$$

The diameter of a proton is 1.68×10^{-15}, which is several orders of magnitude larger than the wavelength of the particle.

Problem 1.2. Using Heisenberg's uncertainty principle, estimate the minimum spread in velocity of an electron confined to a region of size $0.1nm$.

Answer:

Heisenberg's uncertainty principle states that

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}.$$

Rearranging, we find the deviation of the momentum to be

$$\sigma_p \geq \frac{\sigma_x \hbar}{2}.$$

Plugging in our value for σ_x, we get

$$\begin{aligned}\sigma_p &\geq \frac{\sigma_x \hbar}{2} \\ &= \frac{1 \times 10^{-10} \times 1.054 \times 10^{-34}}{2} \\ &= 5.27 \times 10^{-45}.\end{aligned}$$

We can then solve for the velocity to get

$$\begin{aligned}\sigma_p &= \Delta(mv) \\ \Rightarrow \sigma_v &\geq \frac{5.27 \times 10^{-45}}{1 \times 10^{-10}} \\ &\geq 5.27 \times 10^{-35}.\end{aligned}$$

Problem 1.3. Show that for a Gaussian wave packet, the Fourier transform is also Gaussian, and derive $\sigma_k \sigma_x = 1$.

Problem 1.4. Verify that $\psi(\vec{r}, t) = Ae^{i(\vec{k}\cdot\vec{r} - \omega t)}$ solves the free particle Schrödinger equation only if $\hbar\omega = \frac{\hbar^2 k^2}{2m}$.

Answer:

We have the Schrödinger equation to be

$$i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi(\vec{r}, t) + V(\vec{r})\psi(\vec{r}, t).$$

For the free particle case, there are no external interactions with the system, i.e. $V(\vec{r}) = 0$. This gives Schrödinger to be

$$i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi(\vec{r}, t).$$

Plugging in our solution (and assuming a one-dimensioned case), we get

$$i\hbar \frac{\partial}{\partial t}\left[Ae^{i(\vec{k}r - \omega t)}\right] = -\frac{\hbar^2}{2m}\nabla^2\left[Ae^{i(\vec{k}r - \omega t)}\right]. \tag{1}$$

Since we are only considering the 1D case, we have the Laplacian to be $\nabla^2 = \frac{\partial^2}{\partial r^2}$, which gives

$$\begin{aligned}\nabla^2\left[Ae^{i(\vec{k}r - \omega t)}\right] &= \frac{\partial^2}{\partial r^2}\left[Ae^{i(\vec{k}r - \omega t)}\right] \\ &= -k^2 Ae^{i(\vec{k}r - \omega t)}.\end{aligned}$$

Now solving the LHS we get

$$\frac{\partial}{\partial t}\left[Ae^{i(\vec{k}r - \omega t)}\right] = -i\omega Ae^{i(\vec{k}r - \omega t)}.$$

Plugging both of these back into 1, we get

$$\begin{aligned} i\hbar\left[-i\omega A e^{i(\vec{k}r-\omega t)}\right] &= \frac{\hbar^2}{2m}\left[-k^2 A e^{i(\vec{k}r-\omega t)}\right] \\ \hbar\omega A e^{i(\vec{k}r-\omega t)} &= \frac{\hbar^2 k^2}{2m} A e^{i(\vec{k}r-\omega t)} \\ \therefore \hbar\omega &= \frac{\hbar^2 k^2}{2m}. \end{aligned}$$

2 The Mathematical Formalism of Quantum Mechanics

2.1 Hilbert Space

A function is a vector in an infinite-dimensioned vector space. It is sometimes convenient to represent the function using a finite number of "basis functions", so that the vector space necessary to describe the function has finite dimension. This is known as a Hilbert Space.

A vector space requires a suitable basis.

Example 2.1. 3D vector space $\mathbb{R}^3$.

Let $\{\vec{u}_x, \vec{u}_y, \vec{u}_z\}$ be an orthonormal basis in $\mathbb{R}^3$. We can then describe a vector in terms of this basis as

$$\vec{v} = x\vec{u}_x + y\vec{u}_y + z\vec{u}_z,$$

where x, y, and z are coefficients. We then have

$$|\vec{u}_\alpha| = 1, \quad \text{and} \quad \vec{u}_\alpha \cdot \vec{u}_\beta = \delta_{\alpha\beta}.$$

Once we have the basis, we can represent any vector in that basis as a vector of its coefficients in the chosen basis

$$\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

We can denote this vector as $|v\rangle$, indicating it is a normal vector. Similarly, we can define the transpose of this vector as $\langle v|$. Note that we also have to take the complex conjugate of the elements in this vector as we are dealing with complex numbers. We then define the dot product of this to be

$$\vec{a} \cdot \vec{b} \to \vec{a}^T \vec{b} \, \langle a|b\rangle = \sum_{i=1}^{3} a_i^* b_i$$

This is known as Dirac notation.

2.2 Operators

2.2.1 Linear Operators

In a vector space, we can define linear operators (usually matrices) that transform a vector into another

$$|\psi'\rangle = A\,|\psi\rangle\,.$$

Note that $\langle\phi|\psi\rangle$ is a number, and $|\phi\rangle\,\langle\psi|$ is a matrix.

2.2.2 Adjoint Operators

Adjoint operators provide the correspondence between bras and kets

$$|\psi'\rangle = A\,|\psi\rangle \to \langle\psi'| = \langle\psi|\,A^\dagger.$$

Here, $A^\dagger$ is the adjoint of A. It is obtained by transposing and taking the complex conjugate of each element.

2.2.3 Hermitian Operators

If an operator is it's own adjoint (i.e. $A = A^\dagger$), then they are called Hermitian operators. The are simple to recognise

- The elements on opposite side of the diagonal are the complex conjugate of each
- The elements on the diagonal are purely real.

2.3 Basis of a Hilbert Space

A set of kets $\{|\nu_i\rangle\}$ is called orthonormal if

$$\langle \nu_i | \nu_j \rangle = \delta_{ij}.$$

The set $\{|\nu_i\rangle\}$ forms a basis of the Hilbert space ϵ if every $|\psi\rangle$ can be written as

$$|\psi\rangle = \sum_j c_j \, |\nu_j\rangle \, .$$

An important property is the closure relation

$$|\psi\rangle = \sum_j c_j \, |\nu_j\rangle = \sum_j \langle \nu_i | \psi \rangle \, |\nu_j\rangle = \sum_j |\nu_j\rangle \, \langle \nu_k | \psi \rangle \, .$$

For this to be true, it requires

$$\sum_j |\nu_j\rangle \, \langle \nu_j| = \mathbb{I}.$$

This is called the closure relation. The operators $|\nu_j\rangle \, \langle \nu_j|$ projects a vector onto the basis state $|\nu_j\rangle$.

2.4 Observables

General linear algebra theorems

- The eigenvalues of a Hermitian operator are real
- The eigenvectors corresponding to different eigenvalues are orthogonal

A Hermitian operator is called observable if its eigenvectors form a basis of the vector space. We call a unitary an operator for which holds

$$U^{-1} = U^\dagger.$$

The eigenvalues of unitary operators are complex number of modulus 1.

2.5 Problems

3 Spin and The Postulates of Quantum Mechanics

3.1 Postulate 1

The first postulate of quantum mechanics:

First Postulate of Quantum Mechanics 3.1. At an initial time t_0, the state of a quantum system is specified by a ket $|\psi(t_0)\rangle$ that belongs to a suitable vector space ϵ.

This means that any ket

$$|\psi'\rangle = \lambda_1 \, |\psi_1\rangle + \lambda_2 \, |\psi_2\rangle \, ,$$

is also a legitimate state of the system. This is known as the superposition principle.

Note that we normally assume that the state vectors are normalised

$$\langle \psi | \psi \rangle = 1.$$

3.1.1 Application

Assuming a 2D vector space (spin-up and spin-down). We have a basis in this space to be

$$|\uparrow\rangle, |\downarrow\rangle.$$

They must satisfy the following conditions

- Normalisation: $\langle\downarrow|\downarrow\rangle = \langle\uparrow|\uparrow\rangle = 1$
- Orthogonality: $\langle\uparrow|\downarrow\rangle = 0$
- Closure: $|\downarrow\rangle\langle\downarrow| + |\uparrow\rangle\langle\uparrow| = \mathbb{I}$

There are an infinite number of options for this, but the simplest is

$$|\uparrow\rangle = |1\rangle = \begin{pmatrix}1 & 0\end{pmatrix}, \quad |\downarrow\rangle = |2\rangle = \begin{pmatrix}0 & 1\end{pmatrix}.$$

Then, any ket of the form $\alpha|1\rangle + \beta|2\rangle$, with $\alpha^2 + \beta^2 = 1$ is a legitimate state of the system

$$|\psi\rangle = \begin{pmatrix}\alpha & \beta\end{pmatrix}.$$

3.2 Postulate 2

The second postulate of quantum mechanics:

Second Postulate of Quantum Mechanics 3.1. Every measurable physical quantity $\mathcal{A}$ is described by an operator A, which must be an observable.

This means you can construct operators in your space ϵ that not observable, i.e. they cannot correspond to measurable physical quantities.

3.2.1 Application

The physical quantity measured in the Stern-Gerlach experiment is the z-component of the angular momentum $\mathcal{S}_z$. Since it is found to provide only two discrete results, the operator $\mathcal{S}_z$ must be an observable in a 2-dimensional Hilbert space, i.e. a 2×2 matrix.

3.3 Postulate 3

The third postulate of quantum mechanics:

Third Postulate of Quantum Mechanics. When measuring the value of the quantity $\mathcal{A}$, we can only obtain as a result one of the eigenvalues a_n of the observable A.

For an observable $\mathcal{A}$ with a discrete set of eigenvalues a_n, the corresponding physical quantity must be within that set of eigenvalues, not any arbitrary value. This is called quantization.

3.3.1 Application

Using the first two postulates, and the observed values of $\mathcal{S}_z$ of the Stern-Gerlach experiment, we can deduce that S_z must be a 2×2 observable with eigenvalues $\pm\hbar/2$. We can choose to define S_z as the matrix

$$S_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.$$

Proof. We can prove that this is the correct matrix by multiplying them by the observed values

$$S_z |\uparrow\rangle = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}1\\0\end{pmatrix} = \frac{\hbar}{2}|\uparrow\rangle$$
$$S_z |\downarrow\rangle = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix} = -\frac{\hbar}{2}\begin{pmatrix}0\\1\end{pmatrix} = -\frac{\hbar}{2}|\downarrow\rangle,$$

i.e. $|\uparrow\rangle$ and $|\downarrow\rangle$ are eigenvectors of S_z with eigenvalues $\pm\hbar/2$. □

3.4 Postulate 4

Fourth Postulate of Quantum Mechanics. When measuring the quantity $\mathcal{A}$ on a system described by the state $|\psi\rangle$, the probability of obtaining as a result the eigenvalue a_n is given by

$$\boxed{\mathcal{P}(a_n) = |\langle u_n|\psi\rangle|^2}$$

where $|u_n\rangle$ is the eigenvector of A with eigenvalue a_n.

This means the result of the measurement of a quantum system is (generally) probabilistic.

3.4.1 Application

Consider an atom with spin oriented along $\vec{u}$, entering the Stern-Gerlach apparatus. The state of the atom just before the measurement is

$$\left|\psi(t^-)\right\rangle = \cos\frac{\theta}{2}e^{-\frac{i\phi}{2}}|\downarrow\rangle + \sin\frac{\theta}{2}e^{\frac{i\phi}{2}}|\downarrow\rangle = \begin{pmatrix}\cos\frac{\theta}{2}e^{-\frac{i\phi}{2}}\\ \sin\frac{\theta}{2}e^{\frac{i\phi}{2}}\end{pmatrix}.$$

According to the Born's rule (see Def. 3.1),the probability to find the eigenvalue $+\hbar/2$ corresponding to the eigenvector $|\uparrow\rangle$ is

$$\begin{aligned}\mathcal{P}_\uparrow &= \left|\left\langle\uparrow\middle|\psi(t^-)\right\rangle\right|^2\\ &= \left|(1\quad 0)\begin{pmatrix}\cos\frac{\theta}{2}e^{-\frac{i\phi}{2}}\\ \sin\frac{\theta}{2}e^{\frac{i\phi}{2}}\end{pmatrix}\right|^2\\ &= \cos^2\frac{\theta}{2}.\end{aligned}$$

This result is independent of ϕ, since the apparatus only measures the projection along z.

Definition 3.1. Born's Rule

> The Born rule is a postulate of quantum mechanics that gives the probability that a measurement of a quantum system will yield a given result. It states that the probability density for finding a particle at a given position is proportional to the square of the amplitude of the system's wavefunction at that position. (Wikipedia)

It states that an observable that was measured within a quantum system with normalised wavefunction $|\psi\rangle$ corresponds to a self-adjoint operator A whose spectrum is discrete if the measured result will be one of the eigenvalues λ of A, and the probability of measuring a given eigenvalue λ will equal $\langle\psi| P_i |\psi\rangle$, where P_i is the projection onto the eigenspace of A corresponding to λ_i.

For example, a single structureless particle that can be respresented by a wavefunction ψ that depends on position (x, y, z) and time t has the probability density function p for the result of a measurement of the particle at any particular time t_0 to be

$$p(x, y, z, t_0) = |\psi(x, y, z, t_0)|^2.$$

Example 3.2. Take the spin of a particle $\theta = \frac{\pi}{2}$ (spin in the xy-plane). The chance of finding the particle pointing "up" is

$$\mathcal{P}_\uparrow = \cos^2 \frac{pi}{4} = \frac{1}{2}.$$

3.5 Postulate 5

Fifth Postulate of Quantum Mechanics. If the measurement of $\mathcal{A}$ at time t has given the result a_n, immediately after the measurement the system is found in the normalised projection of the state $|\psi(t^-)\rangle$ before the measurement onto the eigenvector $|u_n\rangle$ corresponding to the measured eigenvalue a_n.

$$\left|\psi(t^-)\right\rangle \to \left|\psi(t^+)\right\rangle = \frac{P_n \left|\psi(t^-)\right\rangle}{\sqrt{\left\langle\psi(t^-)\right| P_n \left\langle\psi(t^-)\right|}},$$

where $P_n = |u_n\rangle \langle u_n|$ (Section 3.8) is the projection operator onto the eigenstate $|u_n\rangle$. The denominator normalises the wavefunction.

This is known as the collapse of the wavefunction, and is where taking the measurement of a system "collapses" it into one it's possible states, with the probability of collapsing into this state being defined by the previous Postulate. In general, measuring some property of the system changes its state.

3.5.1 Application

The Stern-Gerlach apparatus can be used to prepare a known quantum state. If we pierce a hole in the screen where the "up" atoms would collide, then we know that those atoms are in the state $|\psi(t^+)\rangle = |\uparrow\rangle$. This new "up" only beam can then be fed into another apparatus to verify this.

Now consider the second apparatus being rotated along $\vec{u}$ instead of purely vertical. Even though we are certain that all the particles are in the state $|\uparrow\rangle$, the outcome of the measurement is not certain. This is because the second apparatus is now measuring $\mathcal{S}_u$ instead of $\mathcal{S}_z$.

We know from Born's rule the probability that each of the eigenvalues may appear as the outcome of the measurement, because we know the quantum state $|\psi\rangle$. Therefore, the average value (called the expectation value) of an observable A given the state $|\psi\rangle$ is

$$\langle A\rangle_\psi = \sum_n a_n \mathcal{P}(a_n) = \sum_n a_n \left|\langle u_n|\psi\rangle\right|^2.$$

Note: given two vectors $|\psi\rangle$ and $|\phi\rangle$, $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$. Factorising further we get

$$\begin{aligned}
&= \sum_n a_n \left|\langle u_n|\psi\rangle\right|^2 \\
&= \sum_n a_n \langle\psi|u_n\rangle \langle u_n|\psi\rangle \\
&= \sum_n \langle\psi| A |u_n\rangle \langle u_n|\psi\rangle \quad \text{note: } a_n |u_n\rangle = A |u_n\rangle \\
&= \langle\psi| A \left[\sum_n |u_n\rangle \langle u_n|\right] |\psi\rangle \quad \text{closure relation} \\
&= \langle\psi| A |\psi\rangle.
\end{aligned}$$

In the example from before:

$$\begin{aligned}
\langle S_u\rangle_{|\uparrow\rangle} &= \langle\uparrow| S_u |\uparrow\rangle \\
&= (1 \quad 0)\frac{\hbar}{2}\begin{pmatrix}\cos\theta & \sin\theta\cdot e^{-i\phi} \\ \sin\theta & -\cos\theta\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} \\
&= \frac{\hbar}{2}\cos\theta.
\end{aligned}$$

3.6 Postulate 6

Sixth Postulate of Quantum Mechanics. If a system is in a state $|\psi(t)\rangle$ at time t, its subsequent time evolution is governed by the Schrödinger equation

$$i\hbar\frac{\partial|\psi(t)\rangle}{\partial t} = \mathcal{H}(t)|\psi(t)\rangle$$

where $\mathcal{H}(t)$ is the Hamiltonian of the system, i.e. the operator that describes its total energy.

Quantum mechanics is for the most part deterministic. Given an initial state, the system evolves in a predictable way, according to a linear differential equation called the Schrödinger equation. The probabilistic aspects enter when you try to measure the state.

3.6.1 Application

We have Schrödinger's equation to be (as above)

$$i\hbar\frac{\partial|\psi(t)\rangle}{\partial t} = \mathcal{H}(t)|\psi(t)\rangle\,,$$

where $\mathcal{H}(t)$ is a matrix, $|\psi\rangle$ is a vector, so the Schrödinger equation is a system of linear differential equations. We can write the formal solution as

$$|\psi(t)\rangle = U(t,t_0)|\psi(t_0)\rangle\,,$$

with the condition

$$U(t_0,t_0) = \mathbb{I}.$$

Substituting $|\psi(t)\rangle = U(t,t_0)|\psi(t_0)\rangle$ into the Schrödinger equation gives

$$i\hbar\frac{d}{dt}U(t,t_0) = \mathcal{H}(t)U(t,t_0).$$

Assuming for simplicity that the Hamiltonian is constant, i.e. $\mathcal{H}(t)\to\mathcal{H}$. The differential equation can then be formally integrated to yield

$$\begin{aligned}U(t,t_0) &= \int_{t_0}^{t} -i\frac{\mathcal{H}}{\hbar}U(t',t_0)\\ &= \exp\left[-i\frac{\mathcal{H}}{\hbar}(t-t_0)\right].\end{aligned}$$

Thus, knowing $|\psi(t_0)\rangle$, the quantum state at later times is given by

$$|\psi(t)\rangle = e^{-i\frac{\mathcal{H}}{\hbar}(t-t_0)}|\psi(t_0)\rangle\,,$$

where

$$\boxed{U(t,t_0) = e^{-i\frac{\mathcal{H}}{\hbar}(t-t_0)}}$$

is the time-evolution operator.

3.7 Spin Matrices

In Section 3.3, we had the S_z matrix which described the spin of a particle. We also have the spin projection along x and y to be

$$S_x = \frac{\hbar}{2}\begin{pmatrix}0&1\\1&0\end{pmatrix},\quad S_y = \frac{\hbar}{2}\begin{pmatrix}0&-i\\i&0\end{pmatrix}.$$

(Note that these are the Pauli matrices σ_x, σ_y, σ_z with a prefactor of $\hbar/2$). These matrices have the eigenvectors

$$|X\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle + |\downarrow\rangle\right) \qquad |-X\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle - |\downarrow\rangle\right)$$
$$|Y\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle + i\,|\downarrow\rangle\right) \qquad |-Y\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle - i\,|\downarrow\rangle\right).$$

Consider an arbitrary unit vector $\vec{u}$ in 3-dimensional space. The observable corresponding to the spin projection along $\vec{u}$ is given by

$$\begin{aligned} S_u &= S_x \sin\theta\cos\phi + S_y \sin\theta\sin\phi + S_z\cos\theta \\ &= \frac{\hbar}{2}\begin{pmatrix} \cos\theta & \sin\theta\cdot e^{-i\phi} \\ \sin\theta\cdot e^{i\phi} & -\cos\theta \end{pmatrix}. \end{aligned}$$

The eigenvectors of this spin projection are then

$$\begin{aligned} |U\rangle &= \cos\frac{\theta}{2}e^{-\frac{i\phi}{2}}\,|\uparrow\rangle + \sin\frac{\theta}{2}e^{\frac{i\phi}{2}}\,|\downarrow\rangle \\ |-U\rangle &= -\sin\frac{\theta}{2}e^{-\frac{i\phi}{2}}\,|\uparrow\rangle + \cos\frac{\theta}{2}e^{\frac{i\phi}{2}}\,|\downarrow\rangle. \end{aligned}$$

3.8 Projection Operator

A projection operator extracts the component of a quantum state lying in a chosen direction or subspace. It is used in Born's rule (Def. 3.1) to define the probability of making a specific measurement, or to transform a vector from one basis to another.

In order for a matrix to be a projection matrix, it must be idempotent ($\hat{P}^2 = \hat{P}$), and if it is also Hermitian is called an orthogonal projection matrix (otherwise it is an oblique projection matrix).

Let $|u_n\rangle$ be an eigenstate in some basis. The projection operator is then given by

$$P_n = |u_n\rangle\langle u_n|.$$

To projection a ket $|\chi\rangle$ onto this eigenstate is given by

$$|\chi\rangle_n = P_n|\chi\rangle = |u_n\rangle\langle u_n|\chi\rangle,$$

which gives the probability of finding $|\chi\rangle$ in the eigenstate $|u_n\rangle$. See Problem 3.1 for a proof that $|u_n\rangle\langle u_n|$ acts a projection operator.

3.9 Exponential of a Matrix

We can use Taylor Expansion to calculate some arbitrary function of a matrix, such as e^A.

$$e^x = \sum_{n=0}^{\infty}\frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots.$$

By analogy

$$e^A = \sum_{n=0}^{\infty}\frac{A^n}{n!} = \mathbb{I} + A + \frac{A^2}{2} + \frac{A^3}{6} + \cdots.$$

Note: if A is diagonal, A^n is also diagonal, and every function of A is diagonal also

$$A = \begin{pmatrix} a_1 & 0 & 0 \\ 0\cdots & 0 & \\ 0 & 0 & a_N \end{pmatrix} \rightarrow \quad e^A = \begin{pmatrix} e^{a_1} & 0 & 0 \\ 0 & \cdots & 0 \\ 0 & 0 & e^{a_N} \end{pmatrix}.$$

In general, if $|\phi\rangle$ is an eigenvector of A with eigenvalue a, then $|\phi\rangle$ is also an eigenvector of $F(A)$ with eigenvalue $F(a)$.

3.10 Hamiltonian Eigenbasis

The Hamiltonian is an observable physical quantity, so it must be described by a matrix whose eigenvectors form a basis of the vector space. If we choose to use the eigenvectors of the Hamiltonian as our basis, it means the Hamiltonian appears as a diagonal matrix

$$\mathcal{H} = \begin{pmatrix} E_1 & 0 & 0 \\ 0 & \cdots & 0 \\ 0 & 0 & E_N \end{pmatrix},$$

where $E_1, \cdots, E_N$ are the eigenvalues (in this case they are "eigenenergies"). Therefore, in the Hamiltonian eigenbasis, the time evolution operator is simply

$$U(t, t_0) = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 & 0 \\ 0 & \cdots & 0 \\ 0 & 0 & e^{-i\frac{E_N}{\hbar}(t-t_0)} \end{pmatrix}.$$

If you can describe the system in the Hamiltonian eigenbasis, the time evolution operator can calculated by hand.

Example 3.1. Spin Precession

Classical description: Consider a magnetic moment $\vec{M} = \gamma\vec{S}$ placed in a uniform magnetic field $\vec{B}$. If $\vec{S}$ and $\vec{B}$ are not parallel, there is a torque $\vec{T}$ acting on the angular momentum

$$\vec{T} = \gamma\vec{S} \times \vec{B}.$$

The derivative of the angular momentum is equal to the torque

$$\frac{d\vec{S}}{dt} = \vec{T} = \gamma\vec{S} \times \vec{B}.$$

We can then define the quantity of angular momentum

$$\omega = -\gamma B.$$

Quantum Description: The energy of a magnetic moment $\vec{M} = \gamma\vec{S}$ placed in a uniform magnetic field $\vec{B}$ is

$$W = -\gamma\vec{S} \cdot \vec{B}.$$

Assume that $\vec{B} \parallel \hat{z}$. Since there are no other relevant contribution to the energy of the system, the Hamiltonian is

$$\begin{aligned} \mathcal{H} &= -\gamma S_z B \\ &= -\gamma B \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \omega \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{\hbar\omega}{2} & 0 \\ 0 & -\frac{\hbar\omega}{2} \end{pmatrix}. \end{aligned}$$

Note that this Hamiltonian is time-independent if the magnetic field is constant (which we are assuming).

This Hamiltonian then has the two eigenenergies

$$E_1 = \frac{\hbar\omega}{2}, \quad E_2 = -\frac{\hbar\omega}{2}.$$

We can then plug these into the time evolution operator

$$U(t,t_0) = e^{-i\frac{\mathcal{H}}{\hbar}(t-t_0)} = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 \\ 0 & e^{-i\frac{E_2}{\hbar}(t-t_0)} \end{pmatrix} = \begin{pmatrix} e^{-i\frac{\omega}{2}(t-t_0)} & 0 \\ 0 & e^{i\frac{\omega}{2}(t-t_0)} \end{pmatrix}.$$

This was simple because we used the eigenstates of the Hamiltonian as our basis. We can then prepare the spin in the symmetric eigenstate of S_x

$$|\psi(0)\rangle = |X\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\rangle + |\downarrow\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}.$$

At a later time t, the state will be

$$|\psi(t)\rangle = U(t,0)\,|\psi(0)\rangle = \begin{pmatrix} e^{-i\frac{\omega}{2}t} & 0 \\ 0 & e^{i\frac{\omega}{2}t} \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t} \end{pmatrix}.$$

To make sense of the results of a wavefunction, we use the expectation value of the wavefunction, i.e. the wavefunction averaged over time. From the previous example, we want to calculate the state along the three spatial axes to see how it evolves over time.

Example 3.2. Calculate the expectation value of the wavefunction in the previous problem.

Starting with the z axis, we have

$$\begin{aligned}\langle S_z\rangle_{\psi(t)} &= \langle\psi(t)|\,S_z\,|\psi(t)\rangle \\ &= \frac{\hbar}{4}\begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t}\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t}\end{pmatrix} \\ &= 0. \end{aligned} \tag{2}$$

Then the x axis,

$$\begin{aligned}\langle S_x\rangle_{\psi(t)} &= \langle\psi(t)|\,S_x\,|\psi(t)\rangle \\ &= \frac{\hbar}{4}\begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t}\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t}\end{pmatrix} \\ &= \frac{\hbar}{4}(e^{i\omega t} + e^{-i\omega t}) = \frac{\hbar}{2}\cos\omega t. \end{aligned} \tag{3}$$

And finally for the y axis

$$\begin{aligned}\langle S_y\rangle_{\psi(t)} &= \langle\psi(t)|\,S_y\,|\psi(t)\rangle \\ &= \frac{\hbar}{4}\begin{pmatrix} e^{i\frac{\omega}{2}t} & e^{-i\frac{\omega}{2}t}\end{pmatrix}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\begin{pmatrix} e^{-i\frac{\omega}{2}t} \\ e^{i\frac{\omega}{2}t}\end{pmatrix} \\ &= \frac{\hbar}{4}(-ie^{i\omega t} + ie^{-i\omega t}) = \frac{\hbar}{2}\sin(\omega t). \end{aligned} \tag{4}$$

From 2, we can see that there is no projection along the vertical (z) axis, meaning it remains within the equatorial (xy) plane. From 3 and 4, we see that the x and y projections oscillate 90 out of phase with frequency ω. This is known as Larmor precession.

3.11 Time Evolution Operator

The time evolution operator $U(t,t_0)$ has some important properties:

- It is a unitary operator

- Its eigenvalues are complex numbers with modulus 1, and it conserves the modulus of any vector it operates on (modulus is the magnitude of the complex number)
- It is time reversible.

3.11.1 Unitary Operators

A unitary operator is a bounded linear operator (a linear transformation between vector spaces that maps bounded sets to bounded sets) on a Hilbert space that satisfies $U^\dagger U = UU^\dagger = \mathbb{I}$, where $U^\dagger$ is the adjoint of U.

Example 3.1. Common Unitary Operators

- The identity function (identity matrix) is trivially a unitary operator
- Rotations in $\mathbb{R}^2$ are the simplest nontrivial example of unitary operators.
- The Fourier operator is a unitary operator (i.e. the operator that performs the Fourier transform with proper normalisation).
- Quantum logic gates are unitary operators, however they are not all Hermitian.

3.11.2 Time Reversibility

For simplicity, take $t_0 = 0$. If we know $U(t,0) = e^{-i\frac{\mathcal{H}}{\hbar}t}$, then

$$U(-t,0) = e^{i\frac{\mathcal{H}}{\hbar}t} = U^{-1}(t,0).$$

This means we can track the system back in time by inverting the time evolution operator. This is especially important for quantum computers as (along with being unitary) this imposes that no information can be lost.

3.12 Commutation

In general, the product of matrices is not commutative. However, we can define the commutator of two operators Λ and Ω as

$$[\Lambda, \Omega] = \Lambda\Omega - \Omega\Lambda.$$

Note that the commutator is itself an operator. When this commutator is 0, we say that the operators commute, so they can be multiplied in any order.

In quantum mechanics, commutation relations have an important meaning. When two observables do not commute, it means the order in which the observables are measured changes the outcomes.

Example 3.1. What is the commutation of S_x and S_z?

Calculating the commutation, we have

$$S_z S_x = \frac{\hbar^2}{4}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \frac{\hbar^2}{4}\begin{pmatrix}0 & 1\\ -1 & 0\end{pmatrix}$$
$$S_x S_z = \frac{\hbar^2}{4}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = \frac{\hbar^2}{4}\begin{pmatrix}0 & -1\\ 1 & 0\end{pmatrix}.$$

The commutator is then

$$[S_z, S_x] = S_z S_x - S_x S_z = \frac{\hbar^2}{4}\begin{pmatrix}0 & 2\\ -2 & 0\end{pmatrix} = i\hbar\frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} = i\hbar S_y.$$

Similarly,

$$[S_x, S_y] = i\hbar S_z$$
$$[S_y, S_z] = i\hbar S_x.$$

This means that in the example where there was two Stern-Gerlach machines, the order in which measurements are made matters.

3.12.1 Ehrenfest Theorem

The expectation value of an observable A at time t is given by

$$\langle A \rangle (t) = \langle \psi(t) | A | \psi(t) \rangle .$$

To see how to evolves with time, we can assume A is time-independent, and differentiate with respect to t to get

$$\frac{d}{dt} \langle \psi(t) | A | \psi(t) \rangle = \left[\frac{d}{dt} \langle \psi(t) | \right] A | \psi(t) \rangle + \langle \psi(t) | A \left[\frac{d}{dt} | \psi(t) \rangle \right] .$$

And from the Schrödinger equation, we have

$$\begin{aligned} \frac{d}{dt} | \psi(t) \rangle = \frac{1}{i\hbar} \mathcal{H} | \psi(t) \rangle \Rightarrow \frac{d}{dt} \langle \psi(t) | &= -\frac{1}{i\hbar} \langle \psi(t) | \mathcal{H} \\ \frac{d}{dt} \langle \psi(t) | A | \psi(t) \rangle = \frac{1}{i\hbar} \langle \psi(t) | A\mathcal{H} - \mathcal{H}A | \psi(t) \rangle \Rightarrow i\hbar \frac{d}{dt} \langle A \rangle &= \langle [A, \mathcal{H}] \rangle . \end{aligned}$$

An observable that commutes with the Hamiltonian is a constant of the motion.

3.13 Pauli Matrices

Let us define the following matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} .$$

These are called the Pauli matrices, and together with the identity matrix $\mathbb{I}$, they form a basis of the vector space of 2×2 matrices. This means that any 2×2 matrix with complex coefficients can be written as

$$A = a_1 \mathbb{I} + a_x \sigma_x + a_y \sigma_y + a_z \sigma_z .$$

Any 2-level system can be "mapped" onto a spin, or equivalently, all quantum mechanics 2-level systems are equivalent. For a "true" spin, the Pauli matrices are proportional to the operators that describe the spin orientation along the 3 axes in real space. There are other 2-level systems for which there is no such meaning.

The three matrices σ_x, σ_y, and σ_z can be grouped into a "vector matrices"

$$\vec{\sigma} = \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix} .$$

This is called vector Pauli matrix. It is useful when calculating the inner product of a real vector in 3 dimensions, with the 3 spin operators.

Example 3.1. The Hamiltonian of a spin in a magnetic field $\vec{B}$ is

$$\mathcal{H} = -\gamma \vec{B} \cdot \vec{S} = -\gamma \frac{\hbar}{2} \vec{B} \cdot \vec{\sigma} = -\gamma \frac{\hbar}{2} \left(B_x \sigma_x + B_y \sigma_y + B_z \sigma_z \right) .$$

3.14 Non-Hermitian Operators

All the operators so far have been Hermitian (except the time evolution operator), and described with physical observables. Now consider the following operators

$$\sigma_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma_- = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

These are clearly non-Hermitian. Therefore, they cannot represent physical observables. But, they are useful tools. For example

$$\sigma_+ |\downarrow\rangle = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |\uparrow\rangle .$$

σ_+ is a "flip up" operator. However,

$$\sigma_+ |\uparrow\rangle = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

applying σ_+ to the spin-up results in the zero-vector, with is not a legitimate state vector as it is not normalised. One thing to note is that these matrices can be created by outer product of the spin-up and spin-down eigenstates with themselves, and act as a projection operator.

3.15 Problems

Problem 3.1. Show that $|\psi\rangle \langle\psi|$ acts as a projection operator, and verify the closure relation

$$\sum |\psi\rangle \langle\psi| = 1$$

for an orthonormal basis.

Answer

Let $\hat{P}_\psi \equiv |\psi\rangle \langle\psi|$. When applied to a ket, the bra must bind to the ket, giving

$$\hat{P}_\psi = |\psi\rangle \langle\psi|\chi\rangle = \langle\psi|\chi\rangle |\psi\rangle .$$

Thus it extracts the component of $|\chi\rangle$ along $|\psi\rangle$. If $|\psi\rangle \perp |\chi\rangle$, then $\langle\psi|\chi\rangle = 0$, so $\hat{P}_\psi |\chi\rangle = 0$.

In order for $\hat{P}_\psi$ to be a projection operator, it must satisfy

$$\hat{P}_\psi^2 = \hat{P}_\psi,$$

i.e. applying the operator to the same ket twice is the same as applying only once. If we assume $|\psi\rangle$ is normalised ($\langle\psi|\psi\rangle = 1$), then

$$\begin{aligned} \hat{P}_\psi^2 &= (|\psi\rangle \langle\psi|) (|\psi\rangle \langle\psi|) \\ &= |\psi\rangle \langle\psi|\psi\rangle \langle\psi| \\ &= |\psi\rangle (1) \langle\psi| \\ &= \hat{P}_\psi. \end{aligned}$$

Now checking for Hermiticity, using $(AB)^\dagger = B^\dagger A^\dagger$, $|\psi\rangle^\dagger = \langle\psi|$, and $\langle\psi|^\dagger = |\psi\rangle$ we get

$$\begin{aligned} \hat{P}_\psi^\dagger &= (|\psi\rangle \langle\psi|)^\dagger \\ &= \left(\langle\psi|^\dagger\right) \left(|\psi\rangle^\dagger\right) \\ &= |\psi\rangle \langle\psi| \\ &= \hat{P}_\psi. \end{aligned}$$

Finally, we can check for orthogonality of basis projectors. Let $\{|i\rangle\}$ be an orthonormal basis, $\langle i|j\rangle = \delta_{ij}$. Define $\hat{P}_i = |i\rangle\langle i|$. Then,

$$\begin{aligned}\hat{P}_i\hat{P}_j &= |i\rangle\langle i|j\rangle|j\rangle \\ &= \delta_{ij}|i\rangle\langle j|.\end{aligned}$$

Hence,

$$\hat{P}_i\hat{P}_j = \begin{cases}\hat{P}_i & i = j \\ 0 & i \neq j\end{cases}.$$

This means projections onto different basis directions do not interfere with each other. We can now use these results to verify the closure relation by having the operator act on an arbitrary ket $|\chi\rangle$

$$\begin{aligned}\left(\sum_i |i\rangle\langle j|\right)|\chi\rangle &= \sum_i |i\rangle\langle i|\chi\rangle \\ &= \sum_i c_i|i\rangle, \quad c_i = \langle i|\chi\rangle.\end{aligned}$$

This is the same as the expansion of $|\chi\rangle$ on the complete orthonormal basis $\{|i\rangle\}$

$$|\chi\rangle = \sum_i c_i|i\rangle.$$

Since the two operators act the same on every ket,

$$\sum_i |i\rangle\langle i| = \mathbb{I}.$$

4 Quantum Mechanics in Real Space

4.1 Schrödinger Equations in Real Space

When we are solving the Schrödinger equation, we generally want to solve it in the form

$$\psi(x,t) = \phi(x)\cdot\chi(t),$$

i.e. we want to separate the spatial and temporal components. This gives the S.E. to be

$$i\hbar\phi(x)\frac{\partial\chi(t)}{\partial t} = -\frac{\hbar^2}{2m}\chi(t)\frac{\partial^2}{\partial x^2}\phi(x) + \chi(t)V(x)\phi(x),$$

and dividing both sides by $\phi(x)\cdot\chi(t)$

$$\frac{i\hbar}{\chi(t)}\frac{\partial\chi(t)}{\partial t} = \frac{1}{\phi(x)}\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\phi(x)\right] + V(x).$$

This gives the S.E. to have the temporal-related components on the LHS, and the spatial related components on the RHS. This can only hold if both sides of the equation are a constant, which we set equal to $\hbar\omega$.

We can define the forms of the wavefunctions as below. Starting with the LHS

$$i\hbar\frac{\partial\chi(t)}{\partial t} = \hbar\omega\cdot\chi(t) \Rightarrow \chi(t) = A\cdot\exp(-i\omega t).$$

And with the RHS

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\phi(x) + V(x)\phi(x) = \hbar\omega\cdot\phi(x).$$

This is just the "real space" version of the TISE

$$\mathcal{H}\phi(x) = E\phi(x) \quad \rightarrow \quad \left[\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\phi(x) = E\phi(x),$$

and therefore the function

$$\psi(x,t) = \phi(x)\exp(-i\omega t)$$

is a solution of the Schrödinger equation, if $\phi(x)$ is a solution of the time-independent part. The function $\psi(x,t)$ written above is a stationary state.

4.2 Stationary States

Whilst the function does contain t in the exponent, it isn't time-dependent as this is just a phase-factor which does not matter in isolation.

Proof. Consider an arbitrary ket $|\psi(t)\rangle = |\psi_0\rangle\, e^{-i\omega t}$ and an observable $\mathcal{A}$. The expectation value of the observable is given by

$$\begin{aligned}\langle A\rangle_\psi &= \langle\psi(t)|\,\mathcal{A}\,|\psi(t)\rangle \\ &= \langle\psi_0|\, e^{i\omega t}\mathcal{A}e^{-i\omega t}\,|\psi_0\rangle \\ &= \langle\psi_0|\, A\,|\psi_0\rangle\,,\end{aligned}$$

which is strictly time-independent. □

Quantum systems can be described using density matrices, which are simply the ket and bra multiplied together, $\rho = |\psi\rangle\langle\psi|$. In this case, the density matrix for $|\psi(t)\rangle$ is

$$\rho = |\psi(t)\rangle\langle\psi(t)| = |\psi_0\rangle\, e^{-i\omega t}e^{i\omega t}\langle\psi_0| = |\psi_0\rangle\langle\psi_0|\,,$$

so the phase factor disappears.

As the S.E. is linear, this means a linear equation of its solutions is also a solution

$$\psi(x,t) = \sum_n c_n\psi_n(x)e^{-i\omega_n t}.$$

This means if we know the stationary states $\psi_n(t)$ and their energies $E_n = \hbar\omega_n$, the problem can be easily solved if the initial state is written as a linear combination of stationary states

$$\psi(x,0) = \sum_n c_n\psi_n(x).$$

When considering the wavefunctions as vectors in an infinite-dimensioned space, this is essentially using a diagonal time-evolution operator.

Example 4.1. Consider a system with potential energy being given by

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise}\end{cases}.$$

The particles cannot access this region outside $(0,a)$ due to the infinite potential required to exceed it. This imposes $\psi = 0$ at and beyond the walls. Note that $\frac{\partial\phi}{\partial x}$ does not need to be continuous. Our general solution (Ansatz) is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}.$$

With our boundary condition at $x = a$, we have

$$\psi(a) = 0 \Rightarrow k_n = \frac{n\pi}{a}, n \in \mathbb{Z}.$$

Therefore, the energy eigenvalues $E_n = \frac{\hbar^2 k_n^2}{2m}$ are

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}.$$

Imposing the normalisation of the wavefunction, $\int |\psi(x)|^2 \, dx = 1$, we get

$$\boxed{\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)}$$

The boundary conditions have the effect of selecting a discrete set of possible momenta, and therefore possible energies, for the particle in the well. The stationary states are standing waves in the well.

4.3 Low-Energy Truncation

Since $\mathcal{H}$ is an observable, its eigenvalues $\psi_n(x)$ form a basis of the Hilbert space. In principle, $n \to \infty$, however in practice this can be truncated to a finite set of N low-energy state, for example becayse the temperature is finite, so the very high-energy states are always unoccupied. Therefore, if the Hilbert space is truncated to a finite dimensioned N, we can write a finite set of basis kets as

$$|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad |\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} \quad |\psi_3\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

where the Hamiltonian is simply

$$\mathcal{H} = \begin{pmatrix} E_1 & 0 & \cdots & 0 \\ 0 & E_2 & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & E_N \end{pmatrix}.$$

Each ket $|\psi_n\rangle$ corresponds to the function of space $\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$.

4.4 Non-Stationary States

A linear superposition of states is also a legitimate state

$$|\psi(t)\rangle = \sum_n c_n(t) \, |\phi_n\rangle \, .$$

Example 4.1. Evaluate

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} \left(|\phi_1\rangle + |\phi_2\rangle\right).$$

Since we are using the basis of the eigenstates of the Hamiltonian, the time-evolution operator is diagonal

$$U(t,0) = \begin{pmatrix} e^{-iE_1 t/\hbar} & 0 & \cdots & 0 \\ 0 & e^{iE_2 t/\hbar} & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & e^{-iE_N t/\hbar} \end{pmatrix}.$$

The time-evolved state is then given by

$$|\psi(t)\rangle = U(t,0)\,|\psi(0)\rangle = \frac{1}{\sqrt{2}}\left(e^{-iE_1t/\hbar}\,|\phi_1\rangle + e^{-iE_2t/\hbar}\,|\phi_2\rangle\right).$$

Multiplying by $e^{iE_1t/\hbar}$, we obtain

$$|\psi(t)\rangle\, e^{iE_1t/\hbar} = \frac{1}{\sqrt{2}}\left(|\phi_1\rangle + e^{-i(E_2-E_1)t/\hbar}\,|\phi_2\rangle\right). \tag{5}$$

Note that the global phase factor on the LHS is completely irrelevant, however the relative phase factor on the RHS is what matters. We can call this ω_{12}, and define it as

$$\omega_{12} = \frac{E_2 - E_1}{\hbar},$$

which we can use in Eq. 5 to get

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(|\phi_1\rangle + e^{-i\omega_{12}t}\,|\phi_2\rangle\right).$$

The corresponding function of real space is then

$$\psi(x,t) = \frac{1}{\sqrt{2}}\left(\phi_1(x) + e^{-i\omega_{12}t}\phi_2(x)\right),$$

which gives the probability to find the particle at point x at time t to be

$$|\psi(x,t)|^2 = \frac{1}{2}\phi_1^2(x) + \frac{1}{2}\phi_2^2(x) + \phi_1(x)\phi_2(x)\cos(\omega_{12}t).$$

5 Quantum Tunnelling

5.1 Particle in a Step-Like Potential

Consider a piecewise constant potential

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & x \geq 0 \end{cases},$$

and a particle with total energy E and momentum $p = \hbar k$, moving from left to right ($x \to \infty$). We have our TISE as

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + (V(x) - E)\right]\phi(x) = 0$$
$$\left[\frac{\partial^2}{\partial x^2} + \frac{2m}{\hbar^2}(E - V(x))\right]\phi(x) = 0$$

In order to properly construct this problem, we have to consider each region separately. Starting with the first region $x < 0$ where $E > V$, we have the particle's total energy greater than the potential energy, meaning there is a kinetic component

$$E - V = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m}.$$

Taking our Ansatz to be

$$\phi(x) = Ae^{ikx} + Be^{-ikx},$$

we know that the particle is going in the $+x$ direction, which means $B = 0$. Now considering the second region $x \geq 0$, this case is forbidden as the kinetic energy component cannot be negative, i.e. the particle would just bounce and reflect against the step in potential. Quantum mechanically, however, we can write

$$V - E = \frac{\hbar^2 \alpha^2}{2m}, \quad \alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar},$$

with α taking the place of $-ik$. This then gives our Ansatz to be

$$\phi(x) = Ce^{-\alpha x} + De^{\alpha x}.$$

The first term $Ce^{-\alpha x}$ describes a wavefunction that decays exponentially as the particle tries to go under the step, called an evanescent wave.

In order to construct the full wavefunction, we must have A, B, C, and D chosen such that $\phi(x)$ and $\frac{\partial \phi}{\partial x}$ are continuous (unless the step is infinitely high). The overall problem can then be solved by writing the appropriate solution in each constant potential region, then matching them at the boundary.

Let's consider, as an example, $E > V$ in both regions

$$\phi_1(x) = A_1 e^{ik_1 x} + A_1' e^{-ik_1 x}$$
$$\phi_2(x) = A_2 e^{ik_2 z} + A_2' e^{-ik_2 x}$$

with

$$k_1 = \frac{\sqrt{2mE}}{\hbar}, \quad k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar},$$

and $A_2' = 0$ since the particle comes from the left. We haven't done the same with A_1' as it may reflect at the step. Matching the conditions at $x = 0$ (the boundary), we get

$$A_1 + A_1' = A_2 \qquad \Rightarrow \phi(x) \text{ is continuous at the boundary}$$
$$ik_1 A_1 - ik_1 A_1' = ik_2 A_2 \qquad \Rightarrow \frac{\partial \phi}{\partial x} \text{ is continuous at the boundary}$$

From this, we can calculate the transmission coefficient T from the amplitude in region II travelling right over the amplitude in region I also travelling right. This gives

$$T \equiv \frac{k_2}{k_1} \left| \frac{A_2}{A_1} \right|^2 = \frac{4k_1 k_2}{(k_1 + k_2)^2}.$$

If we instead take the amplitude travelling left in region I, we get the reflection coefficient R

$$R \equiv \left| \frac{A_1'}{A_1} \right|^2 = 1 - T = 1 - \frac{1k_1 k_2}{(k_1 + k_2)^2}.$$

Note that the condition

$$T + R = 1$$

must always hold, as it enforces the particle existing somewhere in the system. In the classical case, you would expect $R = 0$ because the particle has enough energy to overcome the step, it would just slow down. Instead, it has a non-zero probability of still reflecting back.

5.2 Quantum Tunnelling

Now consider a potential system defined as

$$V(x) = \begin{cases} 0 & x < 0 \\ V_0 & 0 \leq x \leq l \\ 0 & x > l \end{cases}.$$

This gives the system of equations

$$\phi_1(x) = A_1 e^{ik_1 x} + A_1' e^{-ik_1 x}$$
$$\phi_2(x) = A_2 e^{-\alpha x} + A_2' e^{\alpha x}$$
$$\phi_3(x) = A_3 e^{ik_1 x} + A_3' e^{-ik_1 x},$$

with $A_3' = 0$. This can be solved by breaking the problem into two steps, matching the boundary conditions in region I and II at $x = 0$, then regions II and III at $x = l$. This gives

$$T = \frac{16E(V_0 - E)}{V_0^2} e^{-\alpha l}$$
$$= \frac{16E(V_0 - E)}{V_0^2} e^{-l\frac{\sqrt{2m(V_0 - E)}}{\hbar}}.$$

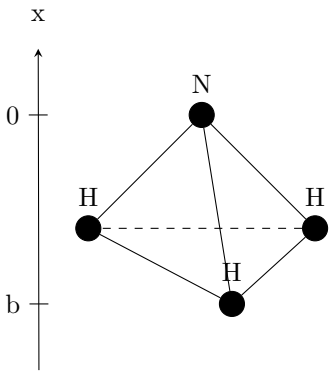
We know that the transmission coefficient $T \neq 0$ because the wavefunction under the barrier is an evanescent wave, so it has an exponentially decaying tail that leaks to the other side of the barrier. This coefficient is proportional to

$$e^{-\alpha l} = e^{-l\frac{\sqrt{2m(V_0 - E)}}{\hbar}},$$

i.e. the probability of tunnelling through the barrier depends exponentially on the width l and on the height of the barrier V_0, measuring from the energy of the incoming particle $V_0 - E$. Also note that the particle can be reflected back with probability $1 - T$, but if it tunnels through the barrier, it comes out with the same energy and momentum as when it arrived. This is called elastic tunnelling.

5.3 Double-Well

An ammonia molecule (NH_3) has an umbrella-like structure



Since the nitrogen is much heavier than the hydrogens, we can think of the nitrogen atom as being fixed at $x = 0$, i.e. the center of mass is effectively where the nitrogen atom is. We will be considering the movement of the three hydrogen atoms. They have two stable positions at $x = \pm b$.

If we want to flip the umbrella upside down, the hydrogen atoms have to overcome the electrostatic repulsion of the nitrogen atom. Thus, there is a potential barrier against flipping the umbrella, but its height is finite. For simplicity, we will approximate the potential as two rectangular wells, with a finite barrier in the middle and infinite walls outside.

5.3.1 Approximation 1: Infinite Barrier

If we approximate the wells as being separated by an infinite potential barrier in the middle, this becomes two independent infinite wells, and each eigenvalue is doubly degenerate. This means that are two orthogonal eigenvectors (one for each well) with the same eigenvalue.

For simplicity we can truncate the Hamiltonian, and assume that $k_B t \ll E_2 - E_1$, so only the 2 lowest energy states $|\phi_{1L}\rangle$, $|\phi_{1R}\rangle$ are populated, which means we have truncated the Hamiltonian to a 2-dimensional

space. We can define our basis as either completely in one well or completely in the other. The two basis states have the same energy E_1, so the Hamiltonian is given by

$$\mathcal{H} = \begin{pmatrix} E_1 & 0 \\ 0 & E_1 \end{pmatrix} = E_1 \mathbb{I}.$$

Since the potential energy is defined with respect to an arbitrary constant, we can set $E_1 = 0$, i.e. measure the energies starting from E_1 upwards. We can also define a different basis, this time with the first ket being positive in both wells, and the second being negative in the first well and positive in the second. We can call these the symmetric and anti-symmetric states, respectively.

$$|\phi_{1S}\rangle = \frac{1}{\sqrt{2}} \left(|\phi_{1R} + |\phi_{1L}\rangle \rangle \right), \quad |\phi_{1A}\rangle = \frac{1}{\sqrt{2}} \left(|\phi_{1R}\rangle - |\phi_{1L}\rangle \right).$$

The states are orthogonal and normalised.

5.3.2 Approximation 2: Finite Wells

Now considering the realistic case, where the barrier between the two wells is not infinite. We know that, in this case, the wavefunction does not go to 0 at the edge, but instead decays exponentially, so it has a finite probability to "leak" to the other side. This is the tunnelling effect. In our symmetric/anti-symmetric basis, both eigenstates now have some leakage under the barrier, but in different amounts, meaning $|\phi_{1S}\rangle$ and $|\phi_{1A}\rangle$ now have different energies. We define $E_A - E_S$ as tunnelling splitting.

5.3.3 Tunnelling Splitting

When the barrier was infinitely high, we had pairs of degenerate states (left/right). With the finite barrier, each pair splits in energy by an amount that depends on how each symmetric/anti-symmetric eigenstate leaks under the barrier, which is tunnelling splitting. This is a measure of the difference in total energy between the two eigenstates for each energy doublet, which increases with higher-energy doublets because they "see" a lower barrier.

5.3.4 Tunnelling Oscillations

We can prepare the system in the initial state

$$|\psi(0)\rangle = \frac{1}{\sqrt{2}} \left(|\phi_{1S}\rangle + |\phi_{1A}\rangle \right) = |\phi_{1R}\rangle .$$

Note that this is the umbrella on the right state, written in the basis of symmetric/anti-symmetric eigenstates of the Hamiltonian. At subsequent times, we'll have

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}} \left(|\phi_{1S}\rangle + e^{-i(E_{1A} - E_{1S})t/\hbar} |\phi_{1A}\rangle \right).$$

This gives the tunnelling oscillations to have the frequency

$$\omega_T = \frac{E_{1A} - E_{1S}}{\hbar}.$$

For ammonia, this is a well defined frequency

$$\omega_T = \frac{E_{1A-E_{1S}}}{\hbar} \approx 2\pi \times^2 4\text{GHz}.$$

5.4 Relation to Spin

In the presence of tunnel coupling, the Hamiltonian of the ammonia molecule (truncated to the lowest-energy doublet) becomes

$$\mathcal{H} = \begin{pmatrix} 0 & -\Delta \\ -\Delta & 0 \end{pmatrix},$$

where

$$2\Delta = E_{1A} - E_{1S}.$$

Tunnel coupling plays the same roles as a magnetic field parallel to x on a spin, which can be seen as the eigenstates $|\phi_{1S}\rangle$ and $|\phi_{1A}\rangle$ look exactly like $|X\rangle$ and $|-X\rangle$ for a spin. The tunnelling oscillations have the same form as the precession in the yz-plane of a spin placed in a magnetic field parallel to x.

Due to the distribution of charge between the atoms, the ammonia molecule as an electric dipole μ, so an electric field applied along the axis of the molecule pulls the umbrella on one side. Therefore, an electric field plays the same role as a magnetic field perpendicular to z, which splits the energy of the natural basis states. Overall, we can write the Hamiltonian as

$$\mathcal{H} = \begin{pmatrix} \epsilon & -\Delta \\ -\Delta & \epsilon \end{pmatrix} = -\Delta\sigma_x + \epsilon\sigma_z,$$

where

$$\begin{cases} 2\Delta = E_{1A} - E_{1S} & \epsilon = 0 \Rightarrow \text{ Tunnel coupling} \\ 2\epsilon = E_{1R} - E_{1L} & \Delta = 0 \Rightarrow \text{ Detuning by electric field} \end{cases}.$$

This means under suitable circumstances (low temperature), a complex system, like an ammonia molecule, is very well described by the same Hamiltonian as that of a spin in a magnetic field that has x and z components.

6 Position and Momentum Operators

6.1 Position Eigenbasis

From Postulate 2 (See Callout 3.1), every physical quantity is described by a Hermitian operator. To do this with position, you would discretise space, for example have a 51-dimensional vector to represent the probability of a particle being in one of 51 positions between $x = 0$ and $x = 1$. We can define a position eigenbasis with orthonormal vectors in this space as

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \cdots \quad |\phi_{51}\rangle = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Each ket represents a particle that is maximally concentrated on the i-th point along the x-axis (in the limit where the discretization step is infinitesimal, it represents a δ- function at the position x_i).

From Postulate 3 (See Callout 3.3), when measuring the value of the quantity $\mathcal{A}$, we can only obtain as a result one of the eigenvalues a_n of the observable $\mathcal{A}$. For example, if we measure X while the state of the system is $|\phi_2\rangle$, we will get $x = 0.02$. So the X operator should look like

$$X = \begin{pmatrix} x_1 & 0 & \cdots & 0 \\ 0 & x_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & x_N \end{pmatrix} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0.02 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & 1 \end{pmatrix}.$$

Unlike with spin and particle in a box, the operator that's diagonal in the chosen basis is not the Hamiltonian of the system.

6.2 Momentum Operator

We have the TISE, which is the eigenvalue problem for the Hamiltonian

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\phi(x) = E\phi(x).$$

Since the kinetic energy is also given by $\frac{p^2}{2m}$, we can interpret this as meaning that the momentum operator must be

$$P = -i\hbar\frac{\partial}{\partial x}.$$

The derivative is a linear operator, so it must be writable as a matrix. As before, we discretise the space, but now we have to calculate the derivative numerically. We can define Δx to be

$$\Delta x = x_{i+1} - x_i,$$

which gives

$$\frac{\partial\phi(x_i)}{\partial x} \approx \frac{1}{2}\left(\frac{\phi(x_{i+1}) - \phi(x_i)}{\Delta x} + \frac{\phi(x_i) - \phi(x_{i-1})}{\Delta x}\right) = \frac{\phi(x_{i+1}) - \phi(x_{i-1})}{2\Delta x}.$$

This is essentially taking the average of the slope immediately before and immediately after the point x_i. This then gives the matrix representation of the momentum operator to be

$$P = -\frac{i\hbar}{2\Delta x}\begin{pmatrix} 0 & 1 & \cdots & 0 \\ -1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & 1 \\ 0 & \cdots & -1 & 1 \end{pmatrix}.$$

With this, we can write explicitly the Hamiltonian of a particle as

$$\mathcal{H} = \frac{P^2}{2m} + V(x),$$

where P is given by the (non-diagonal) matrix above, and $V(x)$ is easy to write because it's just a diagonal matrix with numerical value of the potential at each point x_i. With this, we can calculate the other operators.

6.3 Commutation Relations

One important thing to note is that X and P do not commute. If we calculate brute-force the commutator using the matrices as above, we get

$$[X, P] = XP - PX = \frac{i\hbar}{2}\begin{pmatrix} 0 & 1 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & 1 \\ 0 & \cdots & 1 & 1 \end{pmatrix} \approx i\hbar\mathbb{I}.$$

This represents, in matrix form, the Heisenberg uncertainty principle.

7 Dynamics and Control of a Quantum Bit

For systems under suitable conditions, i.e. minimal outside interference (this is usually low temperature), the system can be described by a Hamiltonian of the form

$$\mathcal{H} = \epsilon\sigma_z + \Delta\sigma_x.$$

This is known as a qubit Hamiltonian. This is a two-level quantum system, and can be used to encode quantum digital information

- Unperturbed case: $\mathcal{H}_0 = \epsilon\sigma_z$, with eigenstates $|\uparrow\rangle, |\downarrow\rangle$, and eigenvalues $\pm\epsilon$
- Coupled case: $\mathcal{H} = \epsilon\sigma_z + \Delta\sigma_x$, with eigenvalues $\lambda_{1,2} = E_{h,l} = \pm\sqrt{\epsilon^2 + \Delta^2}$.

Proof. We can find the eigenstates of a coupled Hamiltonian as

$$\begin{aligned} \det(\mathcal{H} - \lambda\mathbb{I}) &= 0 \\ \Rightarrow (\epsilon - \lambda)(-\epsilon - \lambda) - \Delta^2 &= \lambda^2 - \epsilon^2 - \Delta^2 = 0 \\ \Rightarrow \lambda_{1,2} = E_{h,l} &= \pm\sqrt{\epsilon^2 + \Delta^2}. \end{aligned}$$

□

Consider a qubit Hamiltonian where Δ is constant and $\epsilon = -\infty \to +\infty$. The eigenvectors will be

$$|\phi_h\rangle = \sin\frac{\theta}{2}|\downarrow\rangle + \cos\frac{\theta}{2}|\uparrow\rangle, \quad |\phi_l\rangle = \cos\frac{\theta}{2}|\downarrow\rangle - \sin\frac{\theta}{2}|\uparrow\rangle,$$

where $\tan\theta = \frac{\Delta}{\epsilon}$ for $0 \le \theta \le \pi$. This is known as energy anticrossing.

We can map this onto a spin that is being subjected to a time-variable magnetic field along z, and a constant magnetic field along x, so that the total field makes an angle θ with the z axis. The energy anticrossing occurs because, since $\Delta \neq 0$, the eigenvalues never degenerate.

7.1 Time Evolution

From above, we know that we can write the time evolution operator as $U(t, t_0) = e^{-i\frac{\mathcal{H}}{\hbar}(t-t_0)}$ and that it is diagonal if we use the eigenvectors of the Hamiltonian as basis of the Hilbert space, i.e.

$$U(t, t_0) = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}(t-t_0)} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & e^{-i\frac{E_N}{\hbar}(t-t_0)} \end{pmatrix}.$$

To calculate the time evolution analytically, we must therefore

1. Express the initial state $|\psi(0)\rangle$ in the eigenbasis of the Hamiltonian

$$|\psi(0)\rangle = \sum_n c_n(0)|\phi_n\rangle = \begin{pmatrix} c_1(0) \\ \vdots \\ c_N(0) \end{pmatrix}.$$

2. Apply the time evolution operator and obtain the time-evolved state as

$$|\psi(t)\rangle = \sum_n c_n(0)e^{-i\frac{E_n}{\hbar}t}|\phi_n\rangle = \begin{pmatrix} e^{-i\frac{E_1}{\hbar}t}c_1(0) \\ \vdots \\ e^{-i\frac{E_N}{\hbar}t}c_N(0) \end{pmatrix}$$

Example 7.1. Consider a qubit initially described by the Hamiltonian $\mathcal{H}_0 = \epsilon\sigma_z$ and prepared in its eigenstate $|\uparrow\rangle = |\psi(0^-)\rangle$. At time $t = 0$, switch on the term $\Delta\sigma_x$. Now, $|\psi(0^-)\rangle$ is no longer an eigenstate.

Answer:

1. Express $|\psi(0^-)\rangle = |\uparrow\rangle$ in the new eigenbasis $|\phi_h\rangle, |\phi_l\rangle$:

$$\begin{aligned}\left|\psi(0^-)\right\rangle &= \cos\frac{\theta}{2}|\phi_h\rangle - \sin\frac{\theta}{2}|\phi_l\rangle \\ &= \begin{pmatrix}\cos\frac{\theta}{2} \\ -\sin\frac{\theta}{2}\end{pmatrix}.\end{aligned}$$

2. Time-evolve the coefficients:

$$\begin{aligned}|\psi(t)\rangle &= e^{-i\frac{E_h}{\hbar}t}\cos\frac{\theta}{2}|\phi_h\rangle - e^{-i\frac{E_l}{\hbar}t}\sin\frac{\theta}{2}|\phi_l\rangle \\ &= \begin{pmatrix}e^{-i\frac{E_h}{\hbar}t}\cos\frac{\theta}{2} \\ -e^{-i\frac{E_l}{\hbar}t}\sin\frac{\theta}{2}\end{pmatrix}.\end{aligned}$$

We started in the state $|\uparrow\rangle$. The probability to find the spin in $|\downarrow\rangle$ at time t is $|\langle\downarrow|\psi(t)\rangle|^2$. Express $|\downarrow\rangle$ in the new eigenbasis $|\phi_h\rangle, |\phi_l\rangle$.

$$\begin{aligned}|\downarrow\rangle &= \sin\frac{\theta}{2}|\phi_h\rangle + \cos\frac{\theta}{2}|\phi_l\rangle \\ &= \begin{pmatrix}\sin\frac{\theta}{2} \\ \cos\frac{\theta}{2}\end{pmatrix} \\ \langle\downarrow|\psi(t)\rangle &= \begin{pmatrix}\sin\frac{\theta}{2} & \cos\frac{\theta}{2}\end{pmatrix}\begin{pmatrix}e^{-i\frac{E_h}{\hbar}t}\cos\frac{\theta}{2} \\ -e^{-i\frac{E_l}{\hbar}t}\sin\frac{\theta}{2}\end{pmatrix} \\ &= \sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{-i\frac{E_h}{\hbar}} - \sin\frac{\theta}{2}\cos\frac{\theta}{2}e^{-i\frac{E_l}{\hbar}t} \\ &= \frac{1}{2}\sin\theta\left[e^{-i\frac{E_h}{\hbar}t} - e^{-i\frac{E_l}{\hbar}t}\right].\end{aligned}$$

Note $\sin\theta = \frac{\Delta}{\sqrt{\epsilon^2+\Delta^2}}$.

$$E_h = -E_l = \sqrt{\epsilon^2+\Delta^2} \Rightarrow e^{-i\frac{E_h}{\hbar}t} - e^{-i\frac{E_l}{\hbar}t} = 2i\sin\left(\frac{\sqrt{\epsilon^2+\delta^2}}{\hbar}t\right).$$

$$\boxed{|\langle\downarrow|\psi(t)\rangle|^2 = i\frac{\Delta^2}{\sqrt{\epsilon^2+\Delta^2}}\sin\left(\frac{\sqrt{\epsilon^2+\Delta^2}}{\hbar}t\right).}$$

7.2 Rabi Formula and Oscillation

In Example 7.1, we found the probability of finding a particle in a certain state after applying an external field. Simplifying a bit further, we found

$$\mathcal{P}_\downarrow = |\langle\downarrow|\psi(t)\rangle|^2 = \frac{\Delta^2}{\epsilon^2+\Delta^2}\sin^2\left(\frac{\sqrt{\epsilon^2+\Delta^2}}{\hbar}t\right).$$

This is known as Rabi Formula. The probability oscillates at frequency

$$\omega_R = 2\frac{\sqrt{\epsilon^2+\Delta^2}}{\hbar},$$

which is equal to the energy difference $E_h - E_l$ divided by $\hbar$. This oscillation is known as Rabi Oscillation.

It can be interpreted as a spin starting along z, then precesses around the new total field when B_x is switched on. This also means that we can never get $\mathcal{P}_\downarrow = 1$, as this would require $B_x \to \infty$.

7.3 Time-Dependent Hamiltonian

In certain systems, the Hamiltonian can be explicitly time-dependent (e.g. a spin in a time-varying magnetic field). The time-evolution operator is given by

$$U(t,t_0) = \int_{t_0}^{t} e^{-i\frac{\mathcal{H}(t')}{\hbar}} dt',$$

but the integral is impossible to analytically solve. Instead, we discretise the time in small intervals δt, during which the Hamiltonian is approximated as a constant. Calling $t_n = t_0 + n\delta t$, we get

$$\begin{aligned} U(t_{n+1},t_n) &= e^{-i\frac{\mathcal{H}(t_n)}{\hbar}\Delta t} \\ |\psi(t_{n+1})\rangle &= U(t_{n+1},t_n)\,|\psi(t_n)\rangle \\ |\psi(t_N)\rangle &= \left(\prod_{n=0}^{N-1} U(t_{n+1},t_n)\right)|\psi(t_0)\rangle\,. \end{aligned}$$

This is easy to program, however with small time steps it is very computationally expensive.

Consider a Hamiltonian with a periodic time dependence

$$\mathcal{H}(t) = \mathcal{H}_0 + \mathcal{H}_1\cos(\omega t),$$

with period $T = \frac{2\pi}{\omega}$. Since $\mathcal{H}(t+T) = \mathcal{H}(t)$, we just need to calculate the time evolution operator that takes us from the start to the end of the period, splitting the period T into N small intervals $T = N\Delta t$

$$U(t+T,t) = \prod_{n=0}^{N-1} U(t_{n+1},t_n).$$

$U(t+T,t)$ is called the Floquet operator. We can use this to calculate the state after each period as

$$|\psi(t+kT)\rangle = U^k(t+T,t)\,|\psi(t)\rangle\,.$$

This means we can calculate $U(t+T,t)$ once, then keep multiplying by the same time evolution operator.

7.4 Rotating Frame

As with classical mechanics, a time-dependent Hamiltonian can be transformed into a time-independent one (but not always) by performing a change of basis.

Consider a magnetic dipole $\vec{M}$ placed in a static vertical magnetic field $\vec{B}_0 = B_0\hat{z}$, plus a rotating magnetic field in the xy-plane, $\vec{B}_1 = B_1(\cos\omega t\hat{x} + \sin\omega t\hat{y})$, giving a Hamiltonian of

$$\mathcal{H}(t) = -\vec{M}\cdot\vec{B}(t) = -M_zB_z - M_xB_1\cos\omega t - M_yB_1\sin\omega t.$$

The trick is to define a rotating reference frame $OXYZ$ that rotates at frequency ω around the z-axis. The vector $\vec{B}_1$ appears static in this rotating frame. This is similar to frame of reference in SR.

We can define

- $\omega_0 = \gamma B_0$: Larmor precession frequency of the spin in the static field $\vec{B}_0$.
- $\omega_1 = \gamma B_1$: Larmor frequency the spin would have if it were placed in a static field $\vec{B}_1$.
- $\Delta\omega = \omega - \omega_0$: The "detuning" between precession frequency and rotating field.

This gives our Hamiltonian to be

$$\begin{aligned} \mathcal{H}(t) &= \gamma B_0S_z + \gamma B_1(S_x\cos\omega t - S_y\sin\omega t) \\ &= \omega_0S_z + \omega_1\left(\frac{e^{i\omega t}+e^{-i\omega t}}{2}S_x + \frac{e^{i\omega t}-e^{-i\omega t}}{2i}S_y\right) \\ &= \frac{\hbar}{2}\begin{pmatrix}\omega_0 & \omega_1e^{-i\omega t}\\ \omega_1e^{i\omega t} & -\omega_0\end{pmatrix}. \end{aligned}$$

The generic state at time t in the σ_z basis ($|\uparrow\rangle, |\downarrow\rangle$),

$$|\psi(t)\rangle = c_\uparrow(t)\,|\uparrow\rangle + c_\downarrow(t)\,|\downarrow\rangle = \begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix}.$$

Our time evolution is then given by

$$\begin{aligned} i\hbar\frac{d\,|\psi(t)\rangle}{dt} &= \mathcal{H}(t)\,|\psi(t)\rangle \\ \Rightarrow i\hbar\frac{d}{dt}\begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix} &= \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix}\begin{pmatrix} c_\uparrow(t) \\ c_\downarrow(t) \end{pmatrix} \\ &\begin{cases} i\frac{d}{dt}c_\uparrow(t) = \frac{\omega_0}{2}c_\uparrow(t) + \frac{\omega_1}{2}e^{-i\omega t}c_\downarrow(t) \\ i\frac{d}{dt}c_\downarrow(t) = \frac{\omega_1}{2}e^{i\omega t}c_\uparrow(t) - \frac{\omega_0}{2}c_\downarrow(t) \end{cases} \end{aligned}$$

7.5 Change of Basis

A rotation by an angle ϕ around an axis $\vec{u} = u_x\hat{x} + u_y\hat{y} + u_z\hat{z}$, with $|\vec{u}|^2 = 1$ is expressed in the 2D Hilbert space by the operator

$$R_{\vec{u},\phi} = e^{i\frac{\phi}{2}\sigma_{\vec{u}}},$$

where $\sigma_{\vec{u}} = \vec{u}\cdot\vec{\sigma}$ is the Pauli matrix along the unit vector $\vec{u}$. Note that $R_{\vec{u},\phi}$ is unitary. We change our basis by rotating by ωt around $\hat{z}$, so we need

$$R_{\hat{z},\omega t} = e^{i\frac{\omega t}{2}\sigma_z} = \begin{pmatrix} e^{i\omega t/2} & 0 \\ 0 & e^{-i\omega t/2} \end{pmatrix}.$$

We can define $\left|\tilde{\psi}(t)\right\rangle = R_{\vec{z},\omega t}\,|\psi(t)\rangle = \begin{pmatrix} \tilde{c}_\uparrow(t) \\ \tilde{c}_\downarrow(t) \end{pmatrix}$, which gives

$$\Rightarrow |\psi(t)\rangle = R^{-1}_{\vec{z},\omega t}\left|\tilde{\psi}(t)\right\rangle = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix}\begin{pmatrix} \tilde{c}_\uparrow(t) \\ \tilde{c}_\downarrow(t) \end{pmatrix}.$$

Therefore,

$$\begin{aligned} c_\uparrow(t) = e^{-i\omega t/2}\tilde{c}_\uparrow(t) \Rightarrow \frac{d}{dt}c_\uparrow(t) &= -\frac{i\omega}{2}\tilde{c}(t)e^{-i\frac{\omega t}{2}} + e^{-i\frac{\omega t}{2}}\tilde{c}_\uparrow(t) \\ c_\downarrow(t) = e^{i\omega t/2}\tilde{c}_\downarrow(t) \Rightarrow \frac{d}{dt}c_\downarrow(t) &= \frac{i\omega}{2}\tilde{c}_\downarrow(t)e^{i\frac{\omega t}{2}} + e^{i\frac{\omega t}{2}}\frac{d}{dt}\tilde{c}_\downarrow(t). \end{aligned}$$

We can substitute these expressions for $c_{\uparrow\downarrow}(t)$ and $\frac{dc_{\uparrow\downarrow}(t)}{dt}$ into the Schrödinger equation to get

$$\begin{aligned} i\frac{d}{dt}\tilde{c}_\uparrow(t) &= -\frac{\Delta\omega}{2}\tilde{c}_\uparrow(t) + \frac{\omega_1}{2}\tilde{c}_\downarrow(t) \\ i\frac{d}{dt}\tilde{c}_\downarrow(t) &= \frac{\omega_1}{2}\tilde{c}_\uparrow(t) + \frac{\Delta\omega}{2}\tilde{c}_\downarrow(t). \end{aligned}$$

This is the same as writing

$$i\hbar\frac{d\left|\tilde{\psi}(t)\right\rangle}{dt} = \tilde{\mathcal{H}}\left|\tilde{\psi}(t)\right\rangle,$$

where

$$\tilde{\mathcal{H}} = \frac{\hbar}{2}\begin{pmatrix} -\Delta\omega & \omega_1 \\ \omega_1 & \Delta\omega \end{pmatrix} = -\frac{\hbar}{2}\Delta\omega\sigma_z + \frac{\hbar}{2}\omega_1\sigma_x.$$

In this frame, the Hamiltonian appears time-independent, and all the time-dependence has been moved to the reference frame.

7.6 Magnetic Resonance

The Hamiltonian in the rotating frame looks very similar to a qubit Hamiltonian with

$$\epsilon = -\frac{\hbar}{2}\Delta\omega, \quad \Delta = \frac{\hbar}{2}\omega_1.$$

Assuming the initial state $|\psi(0)\rangle \equiv \left|\tilde{\psi}(t)\right\rangle = |\uparrow\rangle$. The probability to flip the state to $|\downarrow\rangle$ is obtained from the Rabi formula as

$$\mathcal{P}_\downarrow = |\langle\downarrow|\psi(t)\rangle|^2 = \frac{\omega_1^2}{(\Delta\omega)^2 + \omega_1^2}\sin^2\left(\frac{\sqrt{(\Delta\omega)^2+\omega_1^2}}{2}t\right).$$

In the case $\Delta\omega = 0$, i.e. B_1 rotates at the same frequency as the Larmor precession of the spin in the field B_0, then $\mathcal{P}_\downarrow$ can reach 1, which is called magnetic resonance. In the case $\Delta\omega \gg \omega_1, \mathcal{P}_\downarrow \approx 0$, which is called off-resonance, where the spin state is left untouched. At resonances, the Rabi frequency is equal to ω_1, which means the B_0 field is cancelled out in the rotating frame, so the spin precesses at frequency ω_1 around the field B_1.

7.7 Density Operator

A well-defined quantum state, such as the output of a Stern-Gerlach experimentr, is known as pure state as we know with certainty which state it is in. However, in the case that we haven't prepare the input of a Stern-Gerlach machine as a pure state, the output is completely random. Since we can't assign the input state a well-defined ket, the best we can do is describe it as some mixture of $|\uparrow\rangle$ and $|\downarrow\rangle$ with probabilities $\mathcal{P}_\uparrow$ and $\mathcal{P}_\downarrow$, which is known as a mixed state.

For pure states, we can define a density operator, or density matrix as

$$\rho = |\psi\rangle\langle\psi|.$$

Expanding, we have ρ to be

$$\begin{aligned}
|\psi\rangle &= \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix} \\
\Rightarrow \rho &= \begin{pmatrix} c_1 \\ \vdots \\ c_N \end{pmatrix} \begin{pmatrix} c_1^* & \cdots & c_N^* \end{pmatrix} \\
&= \begin{pmatrix} c_1c_1^* & \cdots & c_1c_N^* \\ \vdots & \ddots & \vdots \\ c_Nc_1^* & \cdots & c_Nc_N^* \end{pmatrix},
\end{aligned}$$

or we can call $|u_n\rangle$ the basis vectors of the Hilbert space, we have

$$\rho_{pn} = \langle u_p|\rho|u_n\rangle = c_pc_n^*.$$

The density matrix is Hermitian, and a projector onto the normalised state $|\psi\rangle$ and therefore $\rho = \rho^2$, because

$$\rho^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho.$$

See Problem 3.1 for a proof.

7.7.1 Properties of the Density Operator

- Since $|\psi\rangle$ is normalised, then $\mathrm{Tr}(\rho) = 1$
- Since $\rho = \rho^2$, $\mathrm{Tr}(\rho^2) = 1$

- The expectation value $\langle A\rangle_\psi$ is given by $\langle A\rangle_\psi = \mathrm{Tr}(\rho A)$, since

$$\begin{aligned}\langle A\rangle_\psi &= \langle\psi| A |\psi\rangle \\ &= \sum_{n,p} c_n^* c_p A_{np} \\ &= \sum_{n,p} \langle u_p| \rho |u_n\rangle \langle u_n| A |u_p\rangle \\ &= \sum_p \langle u_p| \rho A |u_p\rangle .\end{aligned}$$

- The time evolution of the density matrix

$$\begin{aligned}\frac{d}{dt}\rho(t) &= \left[\frac{d}{dt}|\psi(t)\rangle\right]\langle\psi(t)| + |\psi(t)\rangle\left[\frac{d}{dt}\langle\psi(t)|\right] \\ &= \frac{1}{i\hbar}\mathcal{H}(t)|\psi(t)\rangle\langle\psi(t)| - \frac{1}{i\hbar}|\psi(t)\rangle\langle\psi(t)|\mathcal{H}(t) \\ &= \frac{1}{i\hbar}[\mathcal{H}(t),\rho(t)] .\end{aligned}$$

- For pure states, the density operator contains the same information as the state vector.

7.7.2 Density Operator for Mixed States

Consider a mixed state. The information we know about the system is that if we measure it in the z-basis, we get $\mathcal{P}_\uparrow = \frac{1}{2} = \mathcal{P}_\downarrow$. We can still write $\langle S_z\rangle = \frac{\hbar}{2}\mathcal{P}_\uparrow - \frac{\hbar}{2}\mathcal{P}_\downarrow = 0$, which would be satisfied if we define

$$\rho = \begin{pmatrix}\mathcal{P}_\uparrow & 0\\ 0 & \mathcal{P}_\downarrow\end{pmatrix},$$

since, using this, we are able to retain the property

$$\langle S_z\rangle = \mathrm{Tr}(\rho S_z).$$

In general, we define the density operator for mixed states as

$$\phi = \begin{pmatrix}\mathcal{P}_1 & \cdots & 0\\ \vdots & \ddots & \vdots\\ 0 & \cdots & \mathcal{P}_N\end{pmatrix},$$

where $\mathcal{P}_n$ is the probability of obtaining the eigenvalue a_n when measuring A. This does not require knowing the precise quantum state at the start of the experiment. Note that it now no longer holds true that $\rho = \rho^2$.

7.7.3 Populations and Coherences

The diagonal elements of ρ are called populations. They describe the probability $\mathcal{P}_n$ of finding the system in the state $|u_n\rangle$ when performing a measurement. This holds for both pure and mixed states.

The off-diagonal elements are called coherences. They describe the phase relationship between different components of a quantum superposition of states.

Example 7.1. We have the two states in the σ_z basis given by

$$|X\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad |Y\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + i|\downarrow\rangle).$$

These correspond to the density matrices

$$|X\rangle \Rightarrow \rho_X = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$
$$|Y\rangle \Rightarrow \rho_Y \frac{1}{2}\begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

The populations are the same, however the coherences are different.

Consider an NMR (nuclear magnetic resonance) experiment, where a large ensemble of identical spins are initially all prepared in the pure state $|X\rangle$. We can describe this ensemble with

$$\rho_X = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

After preparation, each spin precesses at the Larmor frequency $\omega_0 = \gamma B_0$. In reality, each spin sees a slightly different and unknown $B_0(\vec{r})$ field at its location, so after some time we lose track of what exactly is the phase of each spin, and they end up pointing randomly in the xy-plane. We can describe this new ensemble with

$$\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

since we know that each spin is somewhere in the xy-plane, and the coherences have decayed to zero. This phenomenon by which randomness and lack of knowledge of the precise details of the environment of quantum systems results in a loss of knowledge of which pure states the system is in is called decoherence, however it can still be described by a density matrix.

7.7.4 State Purity

Consider the following density matrix

$$\rho_\psi = \frac{1}{2}\begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.25 \\ 0.25 & 0.5 \end{pmatrix},$$

from which we get

$$\rho_\psi^2 = \begin{pmatrix} 0.3125 & 0.25 \\ 0.25 & 0.3125 \end{pmatrix}, \quad \mathrm{Tr}\left(\rho_\psi^2\right) = 0.625 < 1,$$

which shows that ρ_ψ does not describe a pure state. However, it still has nonzero coherences. It means you know something about the phase, however there is some uncertainty. The fully mixed state $\rho = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ would have

$$\mathrm{Tr}(\rho^2) = 0.5,$$

which is less than $\mathrm{Tr}\left(\rho_\psi^2\right)$. $\mathrm{Tr}(\rho^2)$ is called state purity.

8 Coupling and Entanglement of Quantum Particles

8.1 Coupled Quantum Systems

The previous systems were all single-particle systems, and as such they have analogues in the classical world. The more interesting stuff happens when we look at more than one particle.

Consider a spin. It can described by basis state vectors

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

and all its physical properties can be described by suitable combination of Pauli matrices. The way the Pauli matrices act on the basis states is

$$\begin{aligned}
\sigma_x |\uparrow\rangle &= |\downarrow\rangle , & \sigma_x |\downarrow\rangle &= |\uparrow\rangle \\
\sigma_y |\uparrow\rangle &= i |\downarrow\rangle , & \sigma_y |\downarrow\rangle &= -i |\uparrow\rangle \\
\sigma_z |\uparrow\rangle &= |\uparrow\rangle , & \sigma_z |\downarrow\rangle &= - |\downarrow\rangle .
\end{aligned}$$

We can extend these definitions to describe the combined state of two spins. For the basis of our Hilbert space, we can encode the spins using binary, giving

$$|\uparrow\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad |\uparrow\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad |\downarrow\uparrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad |\downarrow\downarrow\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

which is obviously orthonormal. Next, we want to construct the Pauli matrices for each spin. The natural rule to apply is that each matrix should act on its own spin like a normal Pauli matrix, and leave the other spin untouched. Starting with the Pauli-z matrices, they should behave as

$$\sigma_{z1} |\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle , \quad \sigma_{z1} |\uparrow\downarrow\rangle = |\uparrow\downarrow\rangle , \quad \sigma_{z1} |\downarrow\uparrow\rangle = - |\downarrow\uparrow\rangle , \quad \sigma_{z1} |\downarrow\downarrow\rangle = - |\downarrow\downarrow\rangle ,$$

which therefore gives the operator to be

$$\sigma_{z1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Similarly for the other spin, we have

$$\sigma_{z2} |\uparrow\uparrow\rangle = |\uparrow\uparrow\rangle , \quad \sigma_{z2} |\uparrow\downarrow\rangle = - |\uparrow\downarrow\rangle , \quad \sigma_{z2} |\downarrow\uparrow\rangle = |\downarrow\uparrow\rangle , \quad \sigma_{z2} |\downarrow\downarrow\rangle = - |\downarrow\downarrow\rangle ,$$

and therefore

$$\sigma_{z2} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Now with the Pauli-x matrices, we want them to behave as

$$\sigma_{x1} |\uparrow\uparrow\rangle = |\downarrow\uparrow\rangle , \quad \sigma_{x1} |\uparrow\downarrow\rangle = |\downarrow\downarrow\rangle , \quad \sigma_{x1} |\downarrow\uparrow\rangle = |\uparrow\uparrow\rangle , \quad \sigma_{x1} |\downarrow\downarrow\rangle = |\uparrow\downarrow\rangle ,$$

and therefore

$$\sigma_{x1} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Similarly,

$$\sigma_{x1} |\uparrow\uparrow\rangle = |\uparrow\downarrow\rangle , \quad \sigma_{x1} |\uparrow\downarrow\rangle = |\uparrow\uparrow\rangle , \quad \sigma_{x1} |\downarrow\uparrow\rangle = |\downarrow\downarrow\rangle , \quad \sigma_{x1} |\downarrow\downarrow\rangle = |\downarrow\uparrow\rangle ,$$

therefore

$$\sigma_{x2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

8.2 Tensor Product

Rather than constructing these matrices by hand, we can obtain these matrices describing multiple quantum systems by using the tensor product (or Kronecker product). Given two matrices A and B (here 2×2 matrices for simplicity), their tensor product is obtained as the block matrix

$$\begin{aligned} A \otimes B &= \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix} \\ &= \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}. \end{aligned}$$

This is valid for any dimension of matrices being multiplied. If A has dimensions $m \times n$ and B has dimensions $p \times q$, then $A \otimes B$ has $mp \times nq$ dimensions. The tensor product is associative

$$(A \otimes B) \otimes C = A \otimes (B \otimes C),$$

so we just write $A \otimes B \otimes C$. A tensor product can be thought of by replacing each index of matrix A with the entirety of matrix B, whilst multiplying the matrix B by the value previously at the index being replaced.

8.2.1 Construction of a Large Hilbert Space

Now we can use the tensor product to automatically and systematically construct the Hilbert space of as many particles as is necessary. Consider two spins, each one described in the basis

$$|\uparrow\rangle_i = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad |\downarrow\rangle_i \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad i = 1, 2.$$

The basis for their combined Hilbert space is obtained by taking all the combinations of tensor products of individual basis vectors

$$\begin{aligned} |\uparrow\uparrow\rangle &= |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 \\ &= \begin{pmatrix} 1 \times (1 \quad 0)^T \\ 0 \times (1 \quad 0)^T \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned} \qquad \begin{aligned} |\uparrow\downarrow\rangle &= |\uparrow\rangle_1 \otimes |\downarrow\rangle_2 \\ &= \begin{pmatrix} 1 \times (0 \quad 1)^T \\ 0 \times (0 \quad 1)^T \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} |\downarrow\uparrow\rangle &= |\downarrow\rangle_1 \otimes |\uparrow\rangle_2 \\ &= \begin{pmatrix} 0 \times (1 \quad 0)^T \\ 1 \times (1 \quad 0)^T \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \end{aligned} \qquad \begin{aligned} |\downarrow\downarrow\rangle &= |\uparrow\rangle_1 \otimes |\uparrow\rangle_2 \\ &= \begin{pmatrix} 0 \times (0 \quad 1)^T \\ 1 \times (0 \quad 1)^T \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

8.2.2 Operators in a Tensor Product Hilbert Space

Similarly to how we constructed the spin states using the tensor product, we can also obtain the operators by using the same product. To do this, we take the tensor product of the desired operator and identity matrix, with the operator in the position we want affected, and the identity matrix in the position we want left alone.

For example, take the two σ_{xi} operators from before. We can describe them as

$$\sigma_{x1} = \sigma_x \otimes \mathbb{I} = \begin{pmatrix} 0 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 1 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ 1 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & 0 \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\sigma_{x2} = \mathbb{I} \otimes \sigma_x = \begin{pmatrix} 1 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 0 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \times \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

In general, if we have N spins,

$$\sigma_{xi} = \mathbb{I} \otimes \mathbb{I} \otimes \cdots \otimes \sigma_u \otimes \cdots \otimes \mathbb{I},$$

i.e. there are N total products, with the i-th position being the operator.

8.2.3 Product States

In the Hilbert space describing the two spins, we can use the tensor product to write states that represent combinations of individual spin states. For example, to express the state where spin 1 is pointing along $+z$ and spin 2 is pointing along $+x$, we can write

$$|\psi_1\rangle = |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\psi_2\rangle = |X\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The corresponding product state in the combined Hilbert space of the two spins is

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

States obtained from the tensor product of individual particle states are called separable, and they describe the individual properties of separate systems. For example, you can easily verify that

$$\langle\psi| \sigma_{z1} |\psi\rangle = 1, \quad \text{and} \quad \langle\psi| \sigma_{x2} |\psi\rangle = 1$$

as expected.

8.3 Coupled Spins

Consider two electron spins, each described by vector Pauli matrices σ_1, σ_2. When placed in close proximity, they can interact in a variety of ways

- **Dipolar interaction**: each spin produces a magnetic field, which in turn affect the energy of the other. This is analogous to the classical case.
- **Exchange interaction**: the orbits of the electrons overlap, and the quantum mechanical eigenstates of the system are symmetric/antisymmetric combinations of orbitals, to which correspond antisymmetry/symmetric spin configurations.

In any case, the two-spin Hamiltonian can be written in a simplified form as

$$\mathcal{H} = J\sigma_1 \cdot \sigma_2,$$

where $J < 0$ corresponds to a ferromagnetic interaction (favours parallel spin alignment), and $J > 0$ corresponds to an antiferromagnetic interaction (favours antiparallel spin alignment).

We can write out the product of the Pauli matrices

$$\begin{aligned}\mathcal{H} &= J\sigma_1 \cdot \sigma_2 \\ &= J\left(\sigma_{x1}\sigma_{x2} + \sigma_{y1}\sigma_{y2} + \sigma_{z1}\sigma_{z2}\right) \\ &= J\begin{pmatrix}1 & 0 & 0 & 0\\ 0 & -1 & 2 & 0\\ 0 & 2 & -1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}.\end{aligned}$$

The eigenstates of this matrix are

$$\begin{aligned}|T_+\rangle &= \begin{pmatrix}1\\0\\0\\0\end{pmatrix} = |\uparrow\uparrow\rangle \\ |T_0\rangle &= \begin{pmatrix}0\\1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\right) \\ |T_-\rangle &= \begin{pmatrix}0\\0\\0\\1\end{pmatrix} = |\downarrow\downarrow\rangle \\ |S\rangle &= \begin{pmatrix}0\\-1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right).\end{aligned}$$

The first three states are triplet states, with eigenvalue $E_T = J$ (3-fold degenerate), and the last state is a singlet state with eigenvalue $E_S = -3J$ (non-degenerate). This gives the singlet-triplet energy splitting to be $4J$. The factor 4 comes from the definition of the Hamiltonian using Pauli matrices, and if instead it had been defined using spin-$\frac{1}{2}$ matrices, the splitting would be just J.

8.4 Singlet/Triplet States

To understand the meaning of these eigenstates, it's best to calculate the expectation values of some observable. Consider for example the observable that describes the sum of z-projections of the two spins

$$\sigma_{z,tot} = \sigma_{z1} + \sigma_{z2} = \begin{pmatrix}2 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & -2\end{pmatrix}.$$

Using the eigenstates from above, we get

$$\begin{aligned}\langle T_+|\,\sigma_{z,tot}\,|T_+\rangle &= 2 \\ \langle T_0|\,\sigma_{z,tot}\,|T_0\rangle &= 0 \\ \langle T_-|\,\sigma_{z,tot}\,|T_-\rangle &= -2 \\ \langle S|\,\sigma_{z,tot}\,|S\rangle &= 0.\end{aligned}$$

Both $|T_0\rangle$ and $|S\rangle$ have zero total z-projection, meaning that there is nothing differing them from each other in this instance. Instead, consider the square of the total spin

$$\sigma^2 = \left(\sigma_{x1} + \sigma_{x2}\right)^2 + \left(\sigma_{y1} + \sigma_{y2}\right)^2 + \left(\sigma_{z1} + \sigma_{z2}\right)^2 = \begin{pmatrix}8 & 0 & 0 & 0\\ 0 & 4 & 4 & 0\\ 0 & 4 & 4 & 0\\ 0 & 0 & 0 & 8\end{pmatrix}.$$

This gives the expectation values

$$\begin{aligned}\langle T_+|\,\sigma^2\,|T_+\rangle &= 8\\ \langle T_0|\,\sigma^2\,|T_0\rangle &= 8\\ \langle T_-|\,\sigma^2\,|T_-\rangle &= 8\\ \langle S|\,\sigma^2\,|S\rangle &= 0.\end{aligned}$$

All triplet states have the same value of the square of the total spin, whereas the singlet state has zero total spin.

8.5 Entangled States

Consider the singlet state, what is the direction of spin 1 and spin 2 in this state? We can start by writing it out

$$|S\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right) = \frac{1}{\sqrt{2}}\left[\begin{pmatrix}0\\1\\0\\0\end{pmatrix} - \begin{pmatrix}0\\0\\1\\0\end{pmatrix}\right] = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix}.$$

To understand where each spin points, we need to write it as a product state, i.e. find the two 2-dimensional vectors $|\psi_1\rangle$ and $|\psi_2\rangle$ that satisfy

$$|S\rangle = |\psi_1\rangle \otimes |\psi_2\rangle\,,$$

which is impossible. Therefore, $|S\rangle$ and $|T_0\rangle$ are entangled states that cannot be obtained as the tensor product of two single-particle states. It is not possible to attribute a defined state to the individual constituents. This is the meaning of the first postulate of quantum mechanics. For instance, in a 4-dimensional Hilbert space constructed as the tensor product of two 2-dimensional spaces, you can have legitimate states that are not tensor products of states in the 2D spaces.

8.6 Bell's Inequality

Consider two spins, which are prepared initially in some joint state $|\psi\rangle$. We can separate them out, and measure their projection along two axes (defined as the colatitude θ) on each side, α and α' on the left, and β and β' on the right. Express them as Pauli matrices, so that the outcome of the measurement can only be $+1$ or -1. This can repeated multiple times, which allows us to calculate the quantity

$$M = ab - ab' + a'b + a'b' = a(b - b') + a'(b + b').$$

An important note for this is that measurement outcomes can only be ± 1, so it's obvious to see that M can only take values ± 2. Therefore, if the experiment is repeated many times and the average value of M is calculate, we should expect that

$$-2 \leq \langle M\rangle \leq 2.$$

This is known as the Clauser-Horne-Shimony-Holt (CHSH) formulation of Bell's inequality.

8.6.1 Violation of Bell's Inequality

Assume the two spins were initially prepared in a singlet state

$$|S\rangle = \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix}.$$

We wish to calculate the probability $\mathcal{P}_{1,1}$ of obtaining the outcome $a = 1$, $b = 1$ from the measurement of the individual spins along angles α and β. The expression of the eigenstate operator S_u making an angle θ with the vertical axis, with positive eigenvalue

$$|U_\theta\rangle = \cos\frac{\theta}{2}\,|\uparrow\rangle + \sin\frac{\theta}{2}\,|\downarrow\rangle = \begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}\end{pmatrix}.$$

Therefore, the probability is

$$\mathcal{P}_{1,1} = |\langle S|U_\alpha U_\beta\rangle|^2 ,$$

where $|U_\alpha U_\beta\rangle = |U_\alpha\rangle \otimes |U_\beta\rangle$. Expanding this out, we get

$$\begin{aligned}
\mathcal{P}_{1,1} &= |\langle S|U_\alpha U_\beta\rangle|^2 \\
&= \frac{1}{2}\left|(0 \quad 1 \quad -1 \quad 0)\begin{pmatrix} \cos\frac{\alpha}{2}\cos\frac{\beta}{2} \\ \cos\frac{\alpha}{2}\sin\frac{\beta}{2} \\ \sin\frac{\alpha}{2}\cos\frac{\beta}{2} \\ \sin\frac{\alpha}{2}\sin\frac{\beta}{2} \end{pmatrix}\right|^2 \\
&= \frac{1}{2}\left|\cos\frac{\alpha}{2}\sin\frac{\beta}{2} - \sin\frac{\alpha}{2} - \cos\frac{\beta}{2}\right|^2 .
\end{aligned}$$

We have the trigonometric identity

$$\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta,$$

therefore

$$\mathcal{P}_{1,1} = \frac{1}{2}\sin^2\left(\frac{\alpha-\beta}{2}\right).$$

Similarly, we find

$$\begin{aligned}
\mathcal{P}_{-1,-1} &= \frac{1}{2}\sin^2\left(\frac{\alpha-\beta}{2}\right) \\
\mathcal{P}_{1,-1} = \mathcal{P}_{-1,1} &= \frac{1}{2}\cos^2\left(\frac{\alpha-\beta}{2}\right).
\end{aligned}$$

From this, we can calculate the expected average of the outcomes of the measurement of the spins along directions α (on the left) and β (on the right) to be

$$\langle ab\rangle = \mathcal{P}_{1,1} + \mathcal{P}_{-1,-1} - \mathcal{P}_{1,-1} - \mathcal{P}_{-1,1} = -\cos(\alpha-\beta).$$

Also recall that $\cos 2\theta = \cos^2\theta - \sin^2\theta$, and therefore the Bell quantity $\langle M\rangle$ can be rewritten as

$$\langle M\rangle = -\cos(\alpha-\beta) + \cos(\alpha-\beta') - \cos(\alpha'-\beta) - \cos(\alpha'-\beta').$$

We can choose the angles $\alpha, \alpha', \beta, \beta'$ as

$$\begin{aligned}
-\cos(\alpha-\beta) &= -\frac{\sqrt{2}}{2} \\
\cos(\alpha-\beta') &= -\frac{\sqrt{2}}{2} \\
-\cos(\alpha'-\beta) &= -\frac{\sqrt{2}}{2} \\
-\cos(\alpha'-\beta') &= -\frac{\sqrt{2}}{2},
\end{aligned}$$

which we can plug into the above to get

$$\boxed{\langle M\rangle = -2\sqrt{2}}$$

An entangled state can violate Bell's inequality.

8.7 Local Realism

Bell's inequality relies upon three assumptions

- Locality: a physical system cannot be instantly affected by distance objects
- Realism: the physical properties of an object exist already before they are observed
- Free choice: we can freely choose the measurement settings.

To obtain values $\langle M \rangle > 2$ from the inequality, we must admit that either

- Non-Realism: the physical properties of the system which give rise to the values a, a', b, b' do not pre-exist the measurement, and therefore we can't sum the value taken during the different experiments.
- Non-Locality: the choice of measurement axis at one end instantly influences the statistics of the measurement outcomes at the other end
- Super-Determinism: everything, including the action of us setting up a Bell's inequality test and seemingly randomly choosing measurement settings α, α', β, β' was already pre-determined.

8.8 Multi-Particle Density Matrices

Recall the definition of a density matrix: a quantum system described by a state $|\psi\rangle$ can also be described by a density matrix

$$\rho = |\psi\rangle \langle\psi| .$$

Now consider two quantum systems, A and B described by states $|\psi_A\rangle$ and $|\psi_B\rangle$, which are vectors in vector spaces with bases

$$\begin{aligned} \{|u_m\rangle\}, m = 1 \cdots M \quad &\text{for system } A \\ \{|\nu_n\rangle\}, n = 1 \cdots N \quad &\text{for system } B \end{aligned}$$

The combined system will be described by the tensor product state $|\psi_C\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$, which is a state defined in a vector space having a basis $\{|w_p\rangle\}$, the tensor products of $\{|u_m\rangle\}$ and $\{|\nu_n\rangle\}$

$$\{|w_p\rangle\} = \{|u_m\rangle \otimes |\nu_n\rangle\}, m = 1 \cdots M, n = 1 \cdots N \rightarrow p = 1 \cdots M \times N.$$

Note that we will simply write $|u_m\rangle \otimes |\nu_n\rangle$ as $|u_m\rangle |\nu_n\rangle$. It can be shown that the density matrix describing the combined system is just the tensor product of the individual density matrices

$$\rho^C = \rho^A \otimes \rho^B = (|\psi_A\rangle \langle\psi_A|) \otimes (|\psi_B\rangle \langle\psi_B|) .$$

8.9 Partial Trace

Given a tensor-product density matrix $\rho^C = \rho^A \otimes \rho^B$, is it possible to reconstruct the individual ones, ρ^A and ρ^B? Recall that to find the individual matrix elements, we have

$$\begin{aligned} \rho^A_{m,m'} &= \langle u_m | \rho^A | u_{m'} \rangle \\ \rho^B_{n,n'} &= \langle \nu_n | \rho^B | \nu_{n'} \rangle . \end{aligned}$$

Similarly, for the combined system,

$$\rho^C_{p,p'} = \langle w_p | \rho^C | w_{p'} \rangle = \langle u_m | \langle \nu_n | \rho^C | u_{m'} \rangle | \nu_{n'} \rangle .$$

An individual density matrix is recovered by taking the partial trace of the combined density matrix over the basis states of the other particle

$$\begin{aligned} \rho^A_{m,m'} &= \left(\mathrm{Tr}_B \left(\rho^C\right)\right)_{m,m'} = \sum_n \langle u_m | \langle \nu_n | \rho^C | u_{m'} \rangle | \nu_n \rangle \\ \rho^B_{n,n'} &= \left(\mathrm{Tr}_A \left(\rho^C\right)\right)_{n,n'} = \sum_m \langle u_m | \langle \nu_n | \rho^C | u_m \rangle | \nu_{u'} \rangle . \end{aligned}$$

Note the elements that have the same index. The meaning of the partial trace is that it gives the state of one particle, averaged over the possible states of the other.

Example 8.1. Given the two states

$$|\psi_A\rangle = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}, \quad |\psi_B\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

find the density matrices ρ^A, ρ^B, and $\rho^C = \rho^A \otimes \rho^B$, and hence find ρ^B from ρ^C.

We have the density matrices of our two states to be

$$\begin{aligned} |\psi_A\rangle \Rightarrow \rho^A &= \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} \begin{pmatrix} 0.6 & 0.8 \end{pmatrix} \\ &= \begin{pmatrix} 0.36 & 0.48 \\ 0.48 & 0.64 \end{pmatrix}, \\ |\psi_B\rangle \Rightarrow \rho^B &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

From this, we get

$$\begin{aligned} \rho^C &= \rho^A \otimes \rho^B \\ &= \begin{pmatrix} 0.36 & 0 & 0.48 & 0 \\ 0 & 0 & 0 & 0 \\ 0.48 & 0 & 0.64 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Using the partial trace, we get

$$\begin{aligned} \rho^B &= \mathrm{Tr}_A\left(\rho^C\right) \\ &= \left(\langle 0|_A \otimes I_B\right) \rho^C \left(|0\rangle_A \otimes I_B\right) + \left(\langle 1|_A \otimes I_B\right) \rho^C \left(|1\rangle_A \otimes I_B\right) \\ &= \begin{pmatrix} 0.36 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0.64 & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

When a joint state is a product state, averaging over the other has no effect, i.e. there is no influence between the two states.

8.9.1 Partial Trace Over Non-Product States

In the tensor-product vector space, we can find states that are not the tensor product of single-particle states. However, if we use the density matrix ρ instead of the state $|\psi\rangle$, we can always calculate a partial trace. When taking the partial trace over a non-product state, we find that something this is not a pure state, instead it is a mixed state.

For example, the singlet state has a density matrix given by

$$|S\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \rho^S = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The trace over spin A is given by

$$\rho^B = \mathrm{Tr}_A(\rho^S) = \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

Calculating the state purity, we get

$$\mathrm{Tr}\left[\left(\rho^B\right)^2\right] = \mathrm{Tr} \begin{pmatrix} 0.25 & 0 \\ 0 & 0.25 \end{pmatrix} = 0.5 < 1.$$

8.10 Entanglement Entropy

Entropy is a measure of the "disorder" of a system. Consider a physical system that can have N physical states, each with probability p_i of being occupied. The thermodynamic entropy is

$$S = -k_B \sum_i p_i \log p_i.$$

$S = 0$ is only one state $|j\rangle$ is occupied (i.e. $p_i = \delta_{ij}$), and $S = k_B \log N$ is maximum when $p_i = \frac{1}{N}, \quad \forall i$. Shannon entropy extends to this information theory

$$S = \sum_i p_i \log\left(\frac{1}{p_i}\right),$$

where p_i is the probability of obtaining a piece of information. The information gain $\log_2\left(\frac{1}{p_i}\right)$ when we receive the message X_i, and p_i is the probability of receiving it.

The diagonal elements ρ_{ii} of a density matrix describe the probability p_i of finding a system in the state $|u_i\rangle$. Therefore, it is logical to define the entropy of a quantum state as

$$S = -\sum_i p_i \log p_i = -\mathrm{Tr}(\rho \log \rho).$$

For a pure state, $S = 0$. For a maximally mixed state, $\rho = \frac{1}{N}\mathbb{I}$, and $S = \log N$. Recall that the partial trace over the density matrix of an entangled state yields a mixed state. We can therefore use the entropy of the partial trace as a measure of entanglement

$$S = -\mathrm{Tr}\left(\mathrm{Tr}(\rho) \log\left(\mathrm{Tr}_A(\rho)\right)\right).$$

9 Identical Particles in Quantum Mechanics

9.1 Classical and Quantum Identical Particles

In the classical world, fundamental particles (electrons, protons, neutrons, etc) are formally identical. We can assign "labels" to identical particles and write a mathematical state where each particle is identified by some physical property, such as position or velocity. However, we can swap the labels, obtaining a different

mathematical state that corresponds to the same "physical state", since the particles are identical. The physical predictions won't depend on the choice of labels.

In quantum mechanics, it is more complicated. There is no such thing as trajectory, and due to Heisenberg's uncertainty principle, two particles that come in close proximity to each other cannot be differentiated after the interaction. For example, when two particles collide, it is impossible to differentiate the particles once they move away after the interaction.

9.2 Permutation Operator

Consider two spins described by 2×2 Pauli matrices. Since their Hilbert spaces have the same dimension, each one of them can be described in the same basis $\{|u_i\rangle\}, i = 1, 2$. We will use the notation $|u_i(k)\rangle$ for the basis vector $|u_i\rangle$ in the Hilbert space of spin k.

The basis for the combined systems is obtained from the tensor product

$$\{|u_i(1) \otimes |u_j(2)\rangle\rangle\}, \quad i = 1, 2;\ j = 1, 2.$$

Note that $|u_i(1)u_j(2)\rangle$ can be used as a shorthand from the tensor product inside the brackets. The permutation operator $\mathbf{P}_{21}$ is defined by its action on the basis vectors

$$P_{21} |u_i(1)u_j(2)\rangle = |u_i(2)u_j(1)\rangle \equiv |u_j(1)u_i(2)\rangle,$$

which has the effect of swapping the state of spin 2 with that of spin 1. Note that this is true in general, i.e. there is no assumption that the particles are identical.

In general, the tensor product is not commutative, i.e. $A \otimes B \neq B \otimes A$. Therefore, formally, $|u_i(1)u_j(2)\rangle \neq |u_j(2)u_i(1)\rangle$. However, this does not correspond to a different state, it is just the difference in the order of the tensor product used to create the basis for the composite state. Both tensor products describe spin 1 in state $|u_i\rangle$ and spin 2 in state $|u_j\rangle$.

Another thing to note is that $|u_i(1)u_j(2)\rangle \neq |u_j(1)u_i(2)\rangle$ as the particles may not necessarily be identical particles, e.g. an electron and a proton.

9.2.1 Properties of the Permutation Operator

The permutation operator has the following properties

- P_{21} is its own inverse, since $P_{21}^2 = \mathbb{I}$
- $P_{21} = P_{21}^\dagger$
- P_{21} is unitary $P_{21}^\dagger P_{21} = \mathbb{I}$
- The eigenvalues of P_{21} are $+1$ or -1, since their squares are 1
- There are two types of eigenvectors for P_{21}:
  - Symmetric eigenvectors: $P_{21} |\psi_S\rangle = |\psi_S\rangle$
  - Antisymmetric eigenvectors: $P_{12} |\psi_A\rangle = -|\psi_A\rangle$.

Using this operator, we can construct two more operators

$$S = \frac{1}{2}(\mathbb{I} + P_{21}) \qquad \text{symmetrizer}$$

$$A = \frac{1}{2}(\mathbb{I} - P_{21}) \qquad \text{antisymmetrizer}$$

9.3 Symmetrization Postulate

Since we are cannot meaningfully describe indistinguishable particles with pure states, we need an additional postulate for when dealing with multiple particles.

Symmetrisation Postulate. When a system consists of several identical particles, only certain kets can describe the physical state of the system. The physical kets must be either completely symmetric or completely antisymmetric, depending on the nature of the particles.

Particles that require symmetric physical kets are called bosons, and the particles that require antisymmetric ones are called fermions. Bosons therefore have an integer spin, and fermions have half-integer spin.

Using this postulate and properties of the permutation operator, we can define a set of rules to construct the physical kets.

1. Number the particles in a convenient way, and construct the desired ket using tensor products
2. For bosons, apply the symmetrizer operator S, and the antisymmetrizer operator A for fermions.
3. Normalise the resulting ket

9.4 Pauli Exclusion Principle

Consider the cases where the two particles are in the same state, i.e. $|\phi(1)\rangle$ and $|\phi(2)\rangle$. Following the steps from above, we get

1. Tensor product ket: $|\psi\rangle = |\phi(1)\rangle \otimes |\phi(2)\rangle \equiv |\phi(1)\phi(2)\rangle$.
2. For bosons,
$$S|\psi\rangle = \frac{1}{2}(\mathbb{I} + P_{21})|\psi\rangle = \frac{1}{2}(|\phi(1)\phi(2)\rangle + |\phi(2)\phi(1)\rangle) = |\psi(1)\phi(2)\rangle,$$
i.e. we get the simple product tensor of the two states, since it is already symmetric. For fermions,
$$A|\psi\rangle = \frac{1}{2}(\mathbb{I} - P_{21})|\psi\rangle = \frac{1}{2}(|\phi(1)\phi(2)\rangle - |\phi(2)\phi(1)\rangle) = 0,$$
i.e. an absurd result.

This means that two identical fermions cannot occupy the same individual quantum state, which is known as the Pauli exclusion principle.

This is why we need the symmetrisation postulate, because without it we get physically absurd results in certain cases. Take for example two electron spins in a magnetic field $B_0 \parallel z$. Assume the electrons are very close to each other, so that they are unable to be "labelled" by position. The Hamiltonian is of the form

$$\mathcal{H} = \frac{\gamma B_0}{2}(\sigma_{z1} + \sigma_{z2}) = \gamma B_0 \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

in the usual basis. The Hamiltonian is diagonal, and has eigenvalues and vectors

$$\gamma B_0 \to \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\uparrow\rangle, \quad 0 \to \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |\uparrow\downarrow\rangle, \quad 0 \to \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |\downarrow\uparrow\rangle, \quad -\gamma B_0 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Since the 0 eigenstate has two corresponding eigenvectors, it is called doubly-degenerate. These two states have been written as $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$, however this is completely arbitrary, and could be be any of the form

$$|\psi_1\rangle = \alpha_1|\uparrow\downarrow\rangle + \beta_1|\downarrow\uparrow\rangle, \quad |\psi_2\rangle = \alpha_2|\uparrow\downarrow\rangle + \beta_2|\downarrow\uparrow\rangle,$$

as long as $|\alpha_i|^2 + |\beta_i|^2 = 1$ and the two vectors are orthogonal. This seems to indicate that any of these kets constitute a legitimate mathematical description of the physical state, however the predictions on the outcomes of the measurement of the physical observables depend on the choice of coefficients. For example,

we want to pass these spins through a pair of Stern-Gerlach systems, oriented along $+x$. What is the probability of getting $+\frac{\hbar}{2}$? Starting with $|\psi_1\rangle$,the probability of getting $+\frac{\hbar}{2}$ in both measurements is

$$P_{++} = |\langle XX|\psi_1\rangle|^2,$$

where

$$|XX\rangle = |X\rangle \otimes |X\rangle = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}.$$

Therefore,

$$P_{++} = \frac{1}{4}\left|(1 \quad 1 \quad 1 \quad 1)\begin{pmatrix}0\\\alpha_1\\\beta_2\\0\end{pmatrix}\right|^2 = \frac{1}{4}|\alpha_1+\beta_1|^2,$$

which can vary between 0 and $\frac{1}{4}$ depending on the choice of coefficients. This means we cannot freely choose the coefficients of α and β, so this exchange degeneracy must be removed by imposing some specific choice of coefficients. This is what the symmetrisation postulate does, it imposes a choice of coefficients by requiring that the mathematical state obeys a certain symmetry.

9.4.1 Antisymmetrised State

Since electrons are fermions, we need to make an antisymmetric state. The antisymmetriser in 4 dimensions is

$$A = \frac{1}{2}(\mathbb{I} - P_{21}) = \frac{1}{2}\left[\begin{pmatrix}1&0&0&0\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix} - \begin{pmatrix}1&0&0&0\\0&0&1&0\\0&1&0&0\\0&0&0&1\end{pmatrix}\right] = \frac{1}{2}\begin{pmatrix}0&0&0&0\\0&1&-1&0\\0&-1&1&0\\0&0&0&0\end{pmatrix}.$$

Start, for example, from the state $|\uparrow\downarrow\rangle$

$$A|\uparrow\downarrow\rangle = \frac{1}{2}\begin{pmatrix}0&0&0&0\\0&1&-1&0\\0&-1&1&0\\0&0&0&0\end{pmatrix}\begin{pmatrix}0&1&0&0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}0&1&-1&0\end{pmatrix}.$$

Normalising, we get

$$|\psi_A\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&-1&0\end{pmatrix} = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

which is the singlet state. We also need another antisymmetric state with eigenvalue 0, and orthogonal to this one

Proof. We have our eigenstate to be

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Reversing the order of the kets gives

$$-|\psi\rangle = \frac{1}{\sqrt{2}}(|\downarrow\uparrow\rangle - |\uparrow\downarrow\rangle).$$

If we didn't need another antisymmetric state, that would mean that $\uparrow$=$\downarrow$, i.e.

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle - |\uparrow\uparrow\rangle) = 0.$$

This is not a valid state, thus we need two states. This is the Pauli exclusion principle. □

9.4.2 Orbital Wavefunction

An important fact that has so far been omitted is that an electron is not described only by its spin, but also by its position. For example, electrons in adjacent potential wells, with a thin, finite barrier in between, so that their wavefunctions overlap. We can use the basis function $|\phi_L\rangle$ and $|\phi_R\rangle$ to describe the position of each electron. The symmetrisation postulate for fermions says that the total mathematical state must be antisymmetric upon permutation of particles, which must include the orbital component.

From the basis states $|\phi_L\rangle$, $|\phi_R\rangle$, $|\uparrow\rangle$, and $|\downarrow\rangle$, we can construct 16 basis states for the tensor product Hilbert space of the two electrons, which will have basis states like

- $|\phi_L\rangle \otimes |\phi_L\rangle \otimes |\uparrow\rangle \otimes |\uparrow\rangle$: electron 1 on the left with spin up, electron 2 on the left with spin up
- $|\phi_L\rangle \otimes |\phi_R\rangle \otimes |\uparrow\rangle \otimes |\downarrow\rangle$: electron 1 on the left with spin up, electron 2 on the right with spin down, etc.

Note that this gives illegitimate states, they must be antisymmetrical. If the spin part is antisymmetric, then the total wavefunction will be antisymmetric if the orbital part is symmetric, and vice-versa. For example, if the spins are in a singlet state, we want the orbital wavefunction to be the symmetric combination of left and right (ignoring normalisation factor),

$$|\Psi_S\rangle = (|\phi_L\phi_R\rangle + |\phi_R\phi_L\rangle) \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)\,.$$

Or, if the spins are in a triple state (which is spin-symmetric)

$$|\Psi_T\rangle = (|\phi_L\phi_R\rangle - |\phi_R\phi_L\rangle) \otimes (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)\,,$$

which is the second state that we were looking for before.

9.5 Exchange Coupling

In the ammonia molecule from before, the symmetric and antisymmetric orbital combinations have different energy, due to the different amount of wavefunction going under the barrier, once you impose the continuity condition, i.e.

$$E_S < E_A.$$

Since the spin part of the quantum states $(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ and $(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ contributes the same energy (zero), the total states $|\Psi_S\rangle$ and $|\Psi_T\rangle$ differ in energy by $E_A - E_S$, which is called exchange coupling. Note that the states

$$(|\phi_L\phi_R\rangle - |\phi_R\phi_L\rangle) \otimes |\uparrow\uparrow\rangle \quad \text{and} \quad (|\phi_L\phi_R\rangle - |\phi_R\phi_L\rangle) \otimes |\downarrow\downarrow\rangle$$

are also appropriately antisymmetric. In the absence of a magnetic field, we have two sets of states

- Single state: this is given by
$$(|\phi_L\phi_R\rangle + |\phi_R\phi_L\rangle) \otimes (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)\,,$$
and has energy E_S.
- Triplet states: these are given by the tensor product with various states,
$$(|\phi_L\phi_R\rangle - |\phi_R\phi_L\rangle) \otimes \begin{cases} |\uparrow\uparrow\rangle \\ (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \\ |\downarrow\downarrow\rangle \end{cases},$$
which has energy E_A.

9.5.1 Exchange Hamiltonian

From this section, we now know the where the exchange Hamiltonian originates,

$$\mathcal{H} = J\sigma_1 \cdot \sigma_2.$$

Although this appears to only depend on the spin components, the exchange coupling originates from the orbital component of the electrons. Since the singlet-triplet energy splitting is $4J$, we simply have

$$J = \frac{E_A - E_S}{4}.$$

The reason we can write it like this, without explicitly including the orbital degrees of freedom, is because there is a one-to-one relation between orbital and spin states, imposed by the symmetrisation postulate.

10 Atoms, Bonds, and Crystals

10.1 Hydrogen Atom

Using quantum mechanics, we can verify both classical models during the development of the atomic model, as well as see where the postulates arose from. Starting with the crudest picture of the hydrogen atom, we assume that it is a proton with charge $+q$ with an electron with charge $-q$ and mass m_0 orbiting around it at distance r

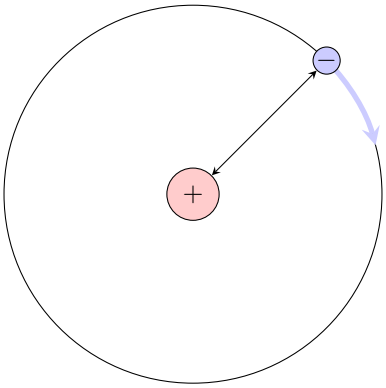


Figure 1: A crude representation of a hydrogen atom. The proton has charge $+q$ and mass m_p, around which orbits at radius r an electron with charge $-q$ and mass m_0. In reality, $m_p \gg m_0$, so we assume the proton is completely stationary.

The Coulomb attractive force between the electron and proton is given by

$$F_q(r) = \frac{q^2}{4\pi\epsilon_0 r^2}.$$

For an equilibrium orbit, the Coulomb attraction must equal the centripetal force

$$\frac{q^2}{4\pi\epsilon_0 r^2} = \frac{m_0 v^2}{r}$$
$$\Rightarrow v = \sqrt{\frac{q^2}{4\pi\epsilon_0 m_0 r}}.$$

The total energy of the atom is the sum of the kinetic energy (assuming the proton doesn't move due to the large relative mass between the electron and proton), and the Coulomb interaction energy

$$E = \frac{1}{2}m_0 v^2 - \frac{q^2}{4\pi\epsilon_0 r} = -\frac{q^2}{8\pi\epsilon_0 r}.$$

The issue with this description is that it does not make any physical sense. If the attraction was gravitational, this would not be an issue, however a charged particle moving in a curved orbit emits electromagnetic energy due to the acceleration, and thus would lose energy and spiral towards the nucleus.

According to the de Broglie hypothesis, every particle has an associated wave with properties of frequency ω and wave number $k = 2\pi/\lambda$ that obey the Planck-Einstein relations

$$\omega = \frac{E}{\hbar}, \quad k = \frac{p}{\hbar}.$$

Since the momentum is $p = m_0 v$ and the kinetic energy is $E_k = \frac{1}{2} m_0 v^2$, the wavelength is

$$\lambda = \frac{h}{m_0 v} = \frac{h}{\sqrt{2 m_0 E_k}}.$$

Using this, we can assert that the stable atomic orbits are where the electron has an integer number of de Broglie wavelengths around the nucleus, i.e.

$$2\pi r = n\lambda = n\frac{h}{m_0 v}.$$

10.2 Bohr Orbits

Combining the orbital quantisation with the velocity of a stable orbit above, we get

$$\begin{aligned} 2\pi r_n &= \frac{nh}{m_0 v} \\ &= \frac{nh}{m_0 q}\sqrt{4\pi\epsilon_0 m_0 r_n} \\ r_n &= n^2 a_0, \end{aligned}$$

where

$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_0 q^2} = 0.529\,\text{Å}$$

is the Bohr radius. From this, the energy of the n-th orbit is

$$E_n = -\frac{q^2}{8\pi\epsilon_0 r_n} = -\frac{1}{n^2} E_0,$$

where

$$E_0 = \frac{q^2}{8\pi\epsilon_0 a_0} = 13.6\,\text{eV}$$

is the Rydberg constant.

10.3 Quantum Treatment

One of the earliest experimental tests of quantum mechanics was the observation that the spectrum of light emitted by hydrogen atoms has discrete frequencies, given by

$$\hbar\omega_{mn} = E_m - E_n = E_0\left(\frac{1}{m^2} - \frac{1}{n^2}\right).$$

We can find the orbital eigenstates of the hydrogen atom by solving the (time-independent) Schrödinger equation

$$\left(-\frac{\hbar^2}{2m_0}\nabla^2 + V(r)\right) + \psi(\vec{r}) = E\psi(\vec{r}),$$

where $V(r)$ is the Coulomb potential. Rewriting in spherical coordinates, we get

$$\nabla^2\psi\left(r,\theta,\phi\right)+\frac{2m_0}{\hbar^2}\left(E-V(r)\right)\psi(r,\theta,\phi)=0.$$

We also want to express the solution in the form

$$\psi(r,\theta,\phi)=R(r)\cdot\Theta(\theta)\cdot\Phi(\phi).$$

Writing the Laplacian in spherical coordinates and expanding the terms, we get

$$\frac{\sin^2\theta}{R}\cdot\frac{\partial}{\partial r}\left(r^2\frac{\partial R}{\partial r}\right)+\frac{1}{\Phi}\cdot\frac{\partial^2\Phi}{\partial\phi^2}+\frac{\sin\theta}{\Theta}\cdot\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\Theta}{\partial\theta}\right)+r^2\sin^2\theta\cdot\frac{2m_0}{\hbar^2}\left(E-V\right)=0.$$

Since we were able to separate the variables, we know that each of these terms must be constant. These constants are the quantum numbers that together completely describe the orbitals of an atom.

The principle quantum number $n=1,2,3,\cdots$ represents the energy level $E_=-\frac{1}{n^2}E_0$, i.e. the eigenstates of the wavefunction. The group of orbitals having the same value of n is called a shell.

The azimuthal quantum number $\ell=0,1,\cdots,n-1$ represents the angular momentum of the electron orbit, in units of $\hbar$. The group of orbitals having the same value of ℓ is called a subshell. It is related to the Schrödinger equation as

$$L^2\psi=\hbar^2\ell\left(\ell+1\right)\psi,$$

where L is the angular momentum operator.

The magnetic quantum number $m_\ell=-\ell,-\ell+1,\cdots,\ell-1,\ell$ represents the z-projection of the angular momentum, and is equal to

$$\frac{1}{\Phi}\cdot\frac{\partial^2\Phi}{\partial\phi^2}=-m^2.$$

Finally, the spin quantum number is just the spin of the electron, $m_s=\pm\frac{1}{2}$.

The orbitals, as defined by ℓ, have 4 usual shapes

- $\ell=0$, s orbitals. This is zero angular momentum, and is spherically symmetric
- $\ell=1$, p orbitals. The angular momentum $L=\hbar$, and its projection along z can be $-\hbar,0,\hbar$ (identified by m_ℓ)
- $\ell=2$, d orbitals. The angular momentum $L=2\hbar$, and its projection along z can be $-2\hbar,-\hbar,0,\hbar,2\hbar$
- $\ell=3$, f orbitals. The angular momentum $L=3\hbar$, and its projection along z can be $-3\hbar,\cdots,3\hbar$.

10.4 Aufbau Principles

We know from the Pauli exclusion principle that no two electrons can occupy the same quantum state. In multi-electron (i.e. everything but hydrogen), the orbitals are filled up starting from the lowest state. Each orbital can take two electrons, in opposite spin state (singlet state). Hund's rule states that all orbitals in a subshell must be occupied before any one orbital is doubly occupied, and all electrons in singly occupied orbitals have the same spin.

10.5 H_2^+ Molecule

Consider two protons and one electron. The protons are separated by distance R. What are the lowest energy states of this system? We can break this into two cases.

10.5.1 Large Distance

If there is a large distance between the protons $R\gg a_0$, then the electron will bind to one of the protons, while the other proton sits by itself. This gives the total energy to be

$$-E_0=-13.6\,\text{eV}.$$

The two states $|1\rangle$ and $|2\rangle$ have the same energy.

10.5.2 Short Distance

If the distance between the protons is relatively short $R \sim a_0$, then the electron can tunnel from one proton to the other. We can call the tunnelling frequency

$$\omega_t = \frac{2\Delta}{\hbar}.$$

In the basis of the state $|1\rangle$ and $|2\rangle$, the Hamiltonian is

$$\mathcal{H} = \begin{pmatrix} -E_0 & -\Delta \\ -\Delta - E_0 & \end{pmatrix} = -E_0\mathbb{I} - \Delta\sigma_x.$$

This gives the eigenstates to be

$$|\phi_S\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right), \quad |\phi_A\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle - |2\rangle\right),$$

and the eigenstates

$$E_S = -E_0 - \Delta, \quad E_A = -E_0 + \Delta.$$

One thing we are missing, however, is the Coulomb repulsion between the protons. In the large distance case, the proton with an electron bound to it will appear as a neutral system to the other proton, so there is no repulsion. In the second case, the electron (in a s-orbital of radius a_0) no longer fully cancels the positive charge, so there start to be a repulsion, which contributes a positive energy $E_r(R)$. By sharing an electron, the system can attain a lower energy than when the electron is located around just one of the protons, which is called a chemical bond. The $|\phi_S\rangle$ state is the bonding orbital, and $|\phi_A\rangle$ is the antibonding orbital.

10.6 Quantum Description of Solids

Solids are formed by placing atoms at regular places in a lattice, therefore their properties need to be understood from the quantum description of atomic orbitals, how they bond and interact, and what happens as a consequence of Pauli exclusion principle and crystal symmetries.

We can consider the simplest possible model of solids, a one-dimensional row of potential wells of width r, with finite barriers. Whilst we can solve this numerically, we need to figure out how many wells are required for an accurate description of a solid.

Rather than solving for a very large number of potential wells, we can instead treat that the first atom in the row is connected to the last (i.e. cyclical). Each level, of energy E_i, is coupled to the same level in the neighbouring well by a tunnel coupling Δ_i, and the last well couples to the first.

Considering first just the first level in each well, we can construct the Hamiltonian with the energies on the diagonal, and the coupling on either off-diagonal,

$$\mathcal{H} = \begin{pmatrix} E_1 & \Delta_1 & 0 & \Delta_1 \\ \Delta_1 & E_1 & \Delta_1 & 0 \\ 0 & \Delta_1 & \ddots & \Delta_1 \\ \Delta_1 & 0 & 0\Delta_1 & E_1 \end{pmatrix}.$$

The transmission coefficient, and thus the tunnel coupling, depends exponentially on the width r of the barrier, and the height measured from the energy of the particle $V_0 - E$,

$$\Delta_1 = k \exp\left(-\frac{\sqrt{2m\left(V_0 - E\right)}}{\hbar} \cdot r\right).$$

Narrowing the barrier results on more leakage of electrons, and the levels split due to the tunnel coupling. We can now add a second level in each well, which gives the Hamiltonian

$$\mathcal{H} = \begin{pmatrix} E_1 & 0 & \Delta_1 & 0 \\ 0 & E_2 & 0 & \Delta_2 \\ \Delta_1 & 0 & E_1 & 0 \\ 0 & \Delta_2 & 0 & E_2 \end{pmatrix}.$$

Since the barrier seen from energy E_2 is lower, we have the new tunnel coupling to be

$$\Delta_2 = k \exp\left(-\frac{\sqrt{2m\left(V_0 - E_2\right)}}{\hbar} \cdot r\right),$$

with $\Delta_2 \gg \Delta_1$ because $(V_0 - E_2) < (V_0 - E_1)$. Because of the stronger tunnel coupling, the states that originate from the second level split faster than the ones from the first level. In addition, these higher-energy orbitals are physically larger, resulting in more of an overlap. Below a certain value of inter-atomic distance, an energy gap appears again, called the band gap.

11 Quantum Transport of Electrical Charge

11.1 Ohm's Law

From a macroscopic view, Ohm's law is given by

$$V = R \cdot I \rightarrow I = G \cdot V,$$

which is equivalent to the local form

$$\vec{J} = \sigma \vec{E},$$

where σ is the conductivity, the reverse of the resistivity ρ. At the microscopic level, the conductivity is expressed as

$$\sigma = \frac{1}{\rho} = \frac{ne^2\tau}{m},$$

where n is the density of charge carriers, e is the electron charge, m is the electron effective mass, and τ is the average time between in-elastic scattering events. This is the Drude model, and it is based on the fact that electric fields accelerates charges, which are then stopped when they hit an obstacle, then accelerate again. The carrier velocity, and therefore current, will depend on the average time between scattering events τ.

We now know that the electrons do not lose their energy by scattering against atoms in a regular crystal lattice. This is because in regular crystal, the quantum mechanical wave function of an electron has the periodicity as the lattice (called Bloch waves). Scattering occurs when you perturb the periodicity, either with an impurity, or with a lattice displacement (thermal vibrations).

11.2 Quantum Transport

If the cross-section of a wire is small enough, quantum effects become important because of wave-particle duality. The maximum energy of the electrons is the Fermi energy E_F, given by

$$E_F = \frac{\hbar^2 k_F^2}{2m} \Rightarrow p_f = \hbar k = \sqrt{2mE_F}$$
$$\lambda_F = \frac{2\pi}{k_F} = \frac{h}{\sqrt{2mE_F}},$$

therefore λ_F is the minimum wavelength of the electrons, and quantum effects arise when the dimensions of the wire approach this wavelength. When the width of the wire is an integer multiple of half this wavelength, it acts as a waveguide, with larger widths carrying more modes. Each mode has a quantum of conductance,

$$G_0 = \frac{e^2}{h} = \frac{1}{25.8\,\mathrm{k}\Omega}.$$

If there are no defects in the wire, the electrons bounce elastically against the walls, so they don't lose any momentum in the direction of the current flow, and all the dissipated power from this conductance is goes into the contacts. The ballistic wire acts as a waveguide, while the contacts act as non-reflecting termination, because an electron that arrived at the drain never goes back into the wire. This conductance is then like the characteristic impedance of the waveguide, and it's non-zero even if the waveguide has no loss.

11.3 Quantum Point Contact

A quantum point contact (QPC) are constrictions of cross-sectional size comparable to the Fermi wavelength, such that quantum effects are visible. Consider a metallic wire, in this case gold. It has Fermi energy given by

$$E_F = 5.53\,\text{eV}, \quad k_F = 1.21 \times 10^{-10}\,\text{m}^{-1} \Rightarrow \lambda_F = 5.2\,\text{Å}$$

The lattice constant in bulk gold is $a = 4.08\,\text{Å}$, so to observe quantised conductance the wire needs to only a couple of atoms thick. When two gold contacts are pulled apart, they spontaneously form a single-atom chain just before breaking. We can clamp a piece of gold wire to a substrate, which is bent and the resistance of the gold wire is monitored until it breaks. The resistance plateaus before breaking, and has conductance $2e^2/h$. The factor of 2 comes because of spin degeneracy, each conductance channel can carry 2 electrons on opposite spin.

11.3.1 2-Dimensional Electron Gas

Different semi-conductors have different band-gaps, which can be used to make heterostructures with interesting energy profiles. A thin layer of electrons is formed at the interface between GaAs and AlGaAs. Metal gates can be fabricated on top of the structure to deplete the 2DEG if required.

11.3.2 QPC in Semiconductors

The Fermi energy in a semiconductor is much smaller than in a metal, giving a larger Fermi wavelength, and QPCs can be made using much larger structures. For example, in the heterostructure from above, you can make a 2-dimensional electron gas with $\lambda_F \sim 20 - 40\text{nm}$, comparable to the size of lithographic features made by electron beam lithography. A negatively biased metal gate on top of the 2DEG is used to deplete the electron layer. With a more negative voltage, the QPC can be narrowed until the channel is completely pinched off, which is equivalent to breaking a wire.

This is much easier to control than a broken wire, however very low temperatures are required to observe clear steps in the conductance. Tuning the QPC on the steep slope between two adjacent steps means that the current is very sensitive to the gate voltage. On this slope, the displacement of just one electron is sufficient to alter the conductance in a measurable way, and is often using quantum-dot structures. It can also detect in real-time an electron jumping between the quantum dot and the large 2-dimensional electron gas.

12 Single-Electron Devices

12.1 Energy-Time Uncertainty Principle

The energy-time uncertainty principle is

$$\Delta E \Delta t \geq \frac{\hbar}{2}.$$

This is different from Heisenberg, as t is a parameter and not a dynamical variable. We define δt from

$$\Delta A = \left| \frac{\langle A \rangle}{t} \right| \delta t,$$

where A is some observable. This means δt represents the typical timescale over which the expectation value of the observable changes in a perceptible way.

Example 12.1. Spin in a magnetic field has a Hamiltonian

$$\mathcal{H} = \omega_0 S_Z + \mathcal{H}'.$$

If we had only the first term $\omega_0 S_Z$, then $|\uparrow\rangle$ and $|\downarrow\rangle$ would be the eigenstates, and the system would stay in these states indefinitely. The extra $\mathcal{H}'$ term describes the fact that if we prepare the high-energy state $|\uparrow\rangle$, after a while it decays to $|\downarrow\rangle$ as it has lower energy. Physically, it is coupled to a

larger system with many states. Since $|\uparrow\rangle$ is no longer an eigenstate of the Hamiltonian, it isn't sharply defined. We say that it has lifetime broadening, meaning that it is an approximate eigenstate with an uncertainty on its energy that relates to its lifetime. Physically, this state is a band of many states that arise from coupling to the larger system, which is what absorbs the energy upon relaxation of $|\uparrow\rangle$ towards $|\downarrow\rangle$.

Example 12.2. Particle in a non-infinite potential well. Since the particle has a non-zero tunnelling probability, it will leave the well after a characteristic time τ. This broadens the energy level of the bound state. The proper way to describe it would be to have a Hamiltonian that couples it to all the states outside the well, which is extremely computationally heavy. Instead, we can approximate it by treating the single-level inside the well, and giving it a broadening $\frac{\hbar}{\tau}$.

12.2 Constants and Conversions

From the Planck-Einstein relations, we already know that energy and frequency are converted into the other by

$$E = \hbar\omega = hf,$$

where $h = 6.64 \times 10^{-37}\,\mathrm{J\,s}$. Energy can also be expressed in temperature units, using the relation

$$E = k_B T,$$

where $k_B = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ is the Boltzmann constant. We also have the unit electron volt eV, the energy of a single electron with charge $e = 1.6 \times 10^{-19}\mathrm{C}$ subjected to a 1 V potential

$$E = e \cdot V.$$

It is then useful to remember the following conversion

$$\boxed{1\,\mathrm{K} = 20.84\,\mathrm{GHz} = 86\,\mu\mathrm{eV}}$$

We can also describe the single-electron charging energy to be

$$E_C = \frac{e^2}{2C}.$$

12.3 Small Capacitors

We can create small electronic devices in the lab using the shadow evaporation, which uses electron beam lithography (EBL) and thermal evaporation of metals

1. Spin on a first layer of polymer resist (MAA)
2. Spin on a "harder" second layer of polymer resist (PMMA) on top
3. Expose pattern with a beam of electrons
4. Put device in a special chemical developer to dissolve resist. The first layer is more sensitive to the exposure, and dissolves quicker
5. Perform first evaporation of aluminium, at an angle. This is the shadow evaporation
6. Oxidise the first aluminium layer, resulting in $\sim 1\,\mathrm{nm}$ thick Al_2O_3
7. Perform a second evaporation at the opposite angle
8. Dissolve the MAA and PMMA resist in acetone

The result is a very small capacitor.

12.3.1 Charging Energy

Consider two layers of aluminium forming. The overlap area is $A = 30 \times 30\text{nm}^2$, and the oxide thickness is $d = 1\,\text{nm}$. If we ignore electron tunnelling, this is just a capacitor with junction capacitance given by

$$C = \epsilon_0 \epsilon_r \frac{A}{d} = 8.85 \times 10^{-12} \times 10 \times \frac{\left(3 \times 10^{-8}\right)^2}{10^{-9}} = 80\,\text{aF}.$$

The charge on a capacitor is $Q = CV$, and the stored energy is

$$E = \frac{1}{2}\frac{Q^2}{C}.$$

Charging the capacitor with a single electron, the charge is $Q = e = 1.6 \times 10^{-19}\,\text{C}$

$$E_C = \frac{e^2}{2C} = \frac{\left(1.6 \times 10^{-19}\right)^2}{1.6 \times 10^{-16}} = 1.6 \times 10^{-22}\,\text{J},$$

which is the charging energy. Dividing by the Boltzmann constant, $k_B = 1.38 \times 10^{-23}\,\text{J}\,\text{K}^{-1}$ to convert to temperature units

$$\frac{E_C}{k_B} = 11.5\,\text{K}.$$

At $T = 1\,\text{K}$, the energy necessary to add a single electron to this capacitor is much larger than the available thermal energy.

12.4 Tunnelling Conductance

Since this oxide barrier is not infinitely high and not infinitely thick, there is a non-zero tunnelling probability, and hence a tunnelling conductance G_T

$$G_T = \frac{1}{R_T} = e^2 N(E_F)\Gamma_T,$$

where $N(E_F)$ is the density of electron states around the Fermi level and $\Gamma_T = \frac{1}{\tau_T}$ is the tunnel rate. This means it will not indefinitely hold charge, and will discharge with a constant

$$\tau = R_T C.$$

Note that this is different from the tunnel time τ_T. The energy uncertainty is then $\Delta E = \frac{\hbar}{\tau} = \frac{\hbar}{R_T C}$. If the resistance approaches the inverse of the quantum conductance, then

$$\lim_{R_T \to \hbar/2e^2} \Delta E = \frac{\hbar}{C}\frac{2e^2}{h} = \frac{e^2}{\pi C},$$

which is comparable to the charging energy

$$\frac{e^2}{2c}.$$

If the tunnelling resistance is of the order to the quantum of resistance, then the energy uncertainty is of the order of the charging energy, and the number of electrons stored in the capacitor is not well defined, and has quantum fluctuations.